

# Translating Solitons of the Mean Curvature Flow Asymptotic to Hyperplanes in $\mathbb{R}^{n+1}$

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A translating soliton is a hypersurface  $M$  in  $\mathbb{R}^{n+1}$  such that the family  $M_t = M - t\mathbf{e}_{n+1}$  is a mean curvature flow, that is, such that normal component of the velocity at each point is equal to the mean curvature at that point  $\mathbf{H} = \mathbf{e}_{n+1}^\perp$ . In this paper we obtain a characterization of hyperplanes that are parallel to the velocity and the family of tilted grim reaper cylinders as the only translating solitons in  $\mathbb{R}^{n+1}$  that are  $C^1$ -asymptotic to two half-hyperplanes outside a non-vertical cylinder. This result was proven for translators in  $\mathbb{R}^3$  by the 2nd author, Perez-Garcia, Savas-Halilaj, and Smoczyk under the additional hypotheses that the genus of the surface was locally bounded and the cylinder was perpendicular to the translating velocity.

## 1 Introduction

Consider  $F(\cdot, t) : M^n \rightarrow \mathbb{R}^{n+1}$  a one-parameter family of smooth immersed hypersurfaces in  $\mathbb{R}^{n+1}$ . We say that  $M_t := F(M, t)$  is a mean curvature flow if it satisfies

$$\begin{aligned}\partial_t F(p, t) &= \mathbf{H}(p, t), \quad p \in M, \quad t > 0, \\ F(p, 0) &= F_0(p),\end{aligned}$$

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Fig. 1. A loop which develops a curve grim reaper.

where  $\mathbf{H}(p, t)$  means the mean curvature vector of the hypersurface  $M_t$  at the point  $F(p, t)$ , and  $F_0$  is a given initial immersion. It is well known (see [29] for instance) that if the initial hypersurface is compact, then the flow develops singularities in finite time. In particular we have that  $|A(p, t)|$  is not bounded when we approach the maximal time  $T$ . Singularities are classified according to the rate at which  $\max_{p \in M_t} |A(p, t)|$  blows up. If there is a constant  $C > 1$  such that

$$\max_{p \in M_t} |A(p, t)| \sqrt{2(T - t)} \leq C,$$

then we say that the flow develops a *Type I singularity* at instant  $T$ . Otherwise, that is, if

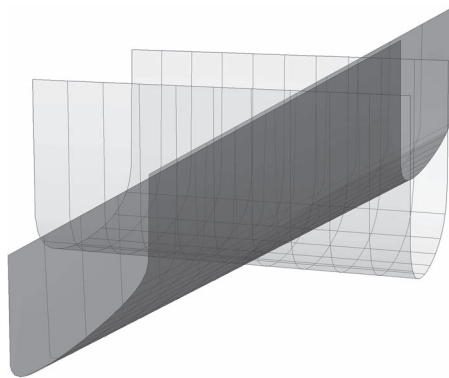
$$\limsup_{t \rightarrow T} \max_{p \in M_t} |A(p, t)| \sqrt{(T - t)} = +\infty,$$

we say that is a *Type II singularity*.

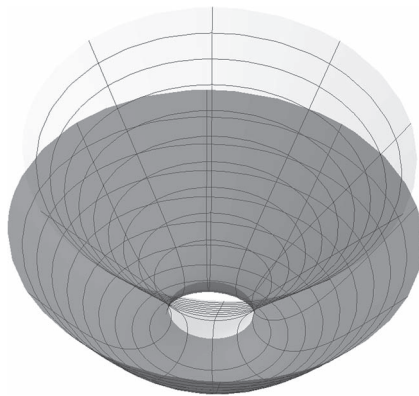
A standard example Type II singularity is given by a loop pinching off to a cusp (see Figure 1). Angenent [3] proved, in the case of convex planar curves, that singularities of this kind are asymptotic (after rescaling) to the grim reaper curve  $y = -\log(\cos x)$ ,  $x \in (-\pi/2, \pi/2)$ , which moves set-wise by translation. In this case, up to inner diffeomorphisms of the soliton, it can be seen as an eternal solution of the curve shortening flow that evolves by translations. In this paper we are interested in this type of solitons, which we will call *translating solitons (or translators)* from now on. In general, a translating soliton is an oriented hypersurface  $M$  in  $\mathbb{R}^{n+1}$  whose mean curvature vector field  $\mathbf{H}$  satisfies  $\mathbf{H} = v^\perp$ , where  $v \in \mathbb{R}^{n+1}$  is a fixed vector. The vector  $v$  is called the velocity. In particular, we have that the scalar mean curvature satisfies

$$H = -\langle v, \xi \rangle, \quad (1)$$

where  $\xi$  is the Gauss map of  $M$ . As we mentioned before, up to an intrinsic diffeomorphism of  $M$ ,  $M_t := M + tv$  is a mean curvature flow. From now on, we will assume (up to dilations and rigid motions) that the velocity of the flow is  $\mathbf{e}_{n+1}$ .



**Fig. 2.** The regular grim reaper cylinder in  $\mathbb{R}^3$  and the tilted grim reaper for  $\theta = \pi/4$ .



**Fig. 3.** The translating catenoid for  $\lambda = 2$ .

The cylinder over a grim reaper curve, that is, the hypersurface in  $\mathbb{R}^{n+1}$  parametrized by  $F : (-\frac{\pi}{2}, \frac{\pi}{2}) \times \mathbb{R}^{n-1} \rightarrow \mathbb{R}^{n+1}$  given by

$$F(x_1, \dots, x_n) = (x_1, \dots, x_n, -\log \cos x_1),$$

is a translating soliton and appears as limit of sequences of parabolic rescaled solutions of mean curvature flows of immersed mean convex hypersurfaces. For example, we can take product of the loop pinching off to a cusp times  $\mathbb{R}^{n-1}$ . We can produce other examples of solitons just by scaling and rotating the grim reaper cylinder.

In this way, we obtain a one-parameter family of translating solitons parametrized by  $F_\theta : \left(-\frac{\pi}{2\cos(\theta)}, \frac{\pi}{2\cos(\theta)}\right) \times \mathbb{R}^{n-1} \longrightarrow \mathbb{R}^{n+1}$

$$F_\theta(x_1, \dots, x_n) = (x_1, \dots, x_n, -\sec^2(\theta) \log \cos(x_1 \cos(\theta)) + \tan(\theta)x_n), \quad (2)$$

where  $\theta \in [0, \pi/2)$  (see Figure 2). Notice that the limit of the family  $F_\theta$ , as  $\theta$  tends to  $\pi/2$ , is a hyperplane parallel to  $\mathbf{e}_{n+1}$ .

Clutterbuck *et al.* [6] (see also [2]) proved that there exists an entire graphical translator in  $\mathbb{R}^{n+1}$  that is rotationally symmetric, strictly convex with translating velocity  $\mathbf{e}_{n+1}$ . This example is known as the translating paraboloid or bowl soliton. Moreover, they classified all the translating solitons of revolution, giving a one-parameter family  $\{W_\lambda^n\}_{\lambda>0}$  of rotationally invariant cylinders called translating catenoids (see Figure 3). The parameter  $\lambda$  controls the size of the neck of each translating soliton. The limit, as  $\lambda \rightarrow 0$ , of  $W_\lambda^n$  consists of two copies of the bowl soliton with a singular point at the axis of symmetry. Furthermore, all these hypersurfaces have the following asymptotic expansion as  $r$  approaches infinity:

$$\frac{r^2}{2(n-1)} - \log r + O(r^{-1}),$$

where  $r$  is the distance function  $\mathbb{R}^n$ .

Recent years have witnessed the appearance of numerous examples of translators; see [7, 13–15] and [20] for more references about construction of translators. It is interesting to notice that all the known examples of complete, properly embedded, translating solitons are asymptotic to either bowl solitons or hyperplanes that are parallel to  $\mathbf{e}_{n+1}$ .

Once this abundance of translating solitons is guaranteed, then arises the need to classify them. One of the 1st classification results was given by Wang in [24]. He characterized the bowl soliton as the only convex translating soliton that is an entire graph. Very recently, Spruck and Xiao [22] have proved that a translating soliton that is a graph over the whole  $\mathbb{R}^2$  must be convex.

Haslhofer [8] showed that any strictly convex, uniformly two-convex translator that is non-collapsing is necessarily rotationally symmetric. In this line of work, Bourni and Langford [4] proved that a translator that arises as a proper blow-up limit of a two-convex mean curvature flow of immersed hypersurfaces is rotationally symmetric.

Using the classic Alexandrov's method of moving hyperplanes, Martín *et al.* [11] showed that the bowl soliton is the only translating soliton that has one end and is

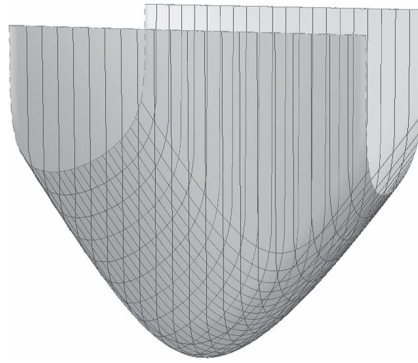
$C^\infty$ -asymptotic to a bowl soliton. Besides that, these authors obtained one of the 1st characterizations of the family of tilted grim reaper cylinders, as the only connected translation solitons in  $\mathbb{R}^{n+1}$ ,  $n \geq 2$ , such that the function  $|A|^2 H^{-2}$  has a local maximum in  $M \setminus H^{-1}(0)$ . Some interesting classification results of grim reaper cylinders can be also found in [23].

Another characterization of the grim reaper cylinder in Euclidean 3-space, in terms of its asymptotic behavior, was given by Martín, Pérez-García, Savas-Halilaj, and Smoczyk [12, 16]. They proved that the grim reaper cylinder is the only connected, properly embedded, translating soliton of dimension 2, with locally bounded genus and being  $C^1$ -asymptotic to two different half-planes. Their proof uses the well-known fact that this kind of solitons can be seen as minimal surfaces in  $\mathbb{R}^3$  with metric  $e^{x_3} \langle \cdot, \cdot \rangle$ , combined with a compactness theorem for minimal surfaces in Euclidean 3-manifolds due to White [28]. This compactness theorem is used for determining the asymptotic shape of the surface. Finally, the authors applied the classic versions of the maximum principle to prove that if a translating soliton is smoothly asymptotic to a grim reaper cylinder, then it must coincide with the grim reaper cylinder.

It is not known whether White's compactness theorem has an extension for higher dimensions and, even in dimension 3, it does not work without the hypothesis of locally bounded genus. So, the proof in [12] fails for higher dimensions and without the hypothesis of locally bounded genus. Moreover, the tilted grim reaper cylinder given by (2) is  $C^1$ -asymptotic to two half-hyperplanes outside a non-horizontal cylinder. Hence, it is natural to ask if it is possible to generalize the theorem for arbitrary dimensions  $n \geq 2$ , without any further assumptions about the topology of the soliton or the axis of the cylinder. Surprisingly, the maximum principle for varifolds and the compactness theorem for stationary integral varifolds allow us to give a positive answer to these questions.

**Theorem 1.1.** The hyperplanes that are parallel to  $\mathbf{e}_{n+1}$  and the family of tilted grim reaper cylinders are the only complete, connected, properly embedded, translating solitons in  $\mathbb{R}^{n+1}$ ,  $n \geq 2$ , which are  $C^1$ -asymptotic to two half-hyperplanes outside a non-vertical solid cylinder. Moreover, if  $n < 7$  hyperplanes parallel to  $\mathbf{e}_{n+1}$  are the only examples that are  $C^1$ -asymptotic to two half-hyperplanes outside a vertical cylinder.

We would like to point out that the previous theorem is sharp in the following sense: if we remove the hypothesis about the cylinder, then the result is false. Given



**Fig. 4.** Ilmanen's example of width  $\sqrt{2}\pi$ .

a vertical slab in  $\mathbb{R}^{n+1}$  of width  $w > \pi$ , Hoffman, Ilmanen, White, and the 2nd author [9] have constructed an  $(n - 1)$ -dimensional family of complete graphical translators over this strip. When the translator that they construct is rotationally symmetric (see also [5]), then these authors also get uniqueness. From now on, we will refer to these examples as  $\Delta$ -wings. When the  $\Delta$ -wing is rotationally symmetric, then it is also convex and it is also asymptotic to the tilted grim reaper cylinder by angle  $\theta = \pm \arccos \frac{\pi}{w}$ . These graphs are asymptotic to the vertical planes  $\{x_1 = \pm w/2\}$ , but it is impossible (by the convexity) to find a cylinder in the hypothesis of our theorem (see Figure 4). On the other hand, if we increase the number of asymptotic planes outside the cylinder, then Nguyen [13, 14] also produced counterexamples.

This paper is structured as follows. In Section 2, we give a short review of results from geometric measure theory that we need in the paper. In Section 3, we obtain a lemma that shows that every complete, properly embedded translating soliton in  $\mathbb{R}^n$  satisfying the assumptions of Theorem 1.1 has a surprising amount of internal dynamical periodicity. Finally, in the last section of this paper, we prove the main theorems.

## 2 Background

In this section we will make a short review of the background we need about translating solitons and geometric measure theory (a short review about varifolds can be found in the appendix of [26]).

## 2.1 Translating solitons

An oriented hypersurface  $M$  in  $\mathbb{R}^{n+1}$  satisfying

$$\mathbf{H} = v^\perp$$

is called a *translating soliton* or *translator* of the mean curvature flow. Recall that we are assuming that  $v = \mathbf{e}_{n+1}$ , where  $B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{n+1}\}$  is the canonical basis of  $\mathbb{R}^{n+1}$ . The following theorem due to Ilmanen gives us the key to relate translating solitons with objects that are very well known and studied—minimal hypersurfaces.

**Theorem 2.1** (Ilmanen [10]). Translating solitons with respect to  $\mathbf{e}_{n+1}$  in  $\mathbb{R}^{n+1}$  are minimal hypersurfaces with respect to the metric  $g = e^{\frac{2}{n}x_{n+1}}\langle \cdot, \cdot \rangle$ , where  $\langle \cdot, \cdot \rangle$  is the Euclidean metric.

Now we are going to define what we mean by a hypersurface being asymptotic to half-hyperplanes outside a cylinder.

**Definition 2.1.** Let  $\mathcal{H}$  a open half-hyperplane in  $\mathbb{R}^{n+1}$  and  $w$  the unit inward pointing normal of  $\partial\mathcal{H}$ . For a fixed positive number  $\delta$ , denote by  $\mathcal{H}(\delta)$  the set given by

$$\mathcal{H}(\delta) := \{p + tw : p \in \partial\mathcal{H} \text{ and } t > \delta\}.$$

We say that a smooth hypersurface  $M$  is  $C^k$ -asymptotic to the open half-hyperplane  $\mathcal{H}$  if  $M$  can be represented as the graph of a  $C^k$ -function  $\varphi : \mathcal{H} \rightarrow \mathbb{R}$  such that for every  $\epsilon > 0$ , there exists  $\delta > 0$ , so that for any  $j \in \{1, 2, \dots, k\}$  it holds

$$\sup_{\mathcal{H}(\delta)} |\varphi| < \epsilon \text{ and } \sup_{\mathcal{H}(\delta)} |D^j \varphi| < \epsilon.$$

We will say that a smooth hypersurface  $M$  is  $C^k$ -asymptotic outside a cylinder to two half-hyperplanes  $\mathcal{H}_1$  and  $\mathcal{H}_2$  if there exists a solid cylinder  $\mathcal{C}$  such that

- i. the solid cylinder  $\mathcal{C}$  contains the boundaries of the half-hyperplane  $\mathcal{H}_1$  and  $\mathcal{H}_2$ ;
- ii.  $M \setminus \mathcal{C}$  consists of two connected components  $M_1$  and  $M_2$  that are  $C^k$ -asymptotic to  $\mathcal{H}_1$  and  $\mathcal{H}_2$ , respectively.

**Remark 2.1.** Notice that the solid cylinders in  $\mathbb{R}^{n+1}$  that we are considering are those whose boundary is isometric to  $\mathbb{S}^1(r) \times \mathbb{R}^{n-1}$ , where  $\mathbb{S}^1(r)$  is the sphere of radius  $r$ .

Simple examples of this case are the hyperplanes parallel to  $\mathbf{e}_{n+1}$ . Other examples are given by the family of tilted grim reaper cylinders, which are  $C^\infty$ -asymptotic to two half-hyperplanes outside the corresponding tilted cylinder.

## 2.2 Maximum principle

In this short section, we are going to give a short review of the maximum principle for varifolds. Before stating it, let us recall the classical version of the maximum principle.

**Theorem 2.2** (Interior Maximum Principle). Let  $M_1$  and  $M_2$  be  $n$ -dimensional embedded translating solitons, with not necessarily empty boundaries, in the Euclidean space  $\mathbb{R}^{n+1}$ . Suppose that there exists a common point  $x$  in the interior of  $M_1$  and  $M_2$ , where the corresponding tangent spaces coincide and that  $M_1$  lies at one side of  $M_2$ . Then  $M_1 = M_2$ .

Although this maximum principle has very useful applications, it requires a very strong hypothesis: smoothness. As we will work with varifolds, we will need another version of this theorem that can be applied in this setting. The version that we will use in this paper is due to Solomon and White in [21] (see also [27, Theorem 4]). Recall that a varifold  $V$  in a Riemannian manifold  $N$  minimizes area to 1st order if the 1st variation of  $V$ , denoted by  $\delta V$ , satisfies

$$\delta V(X) \geq 0,$$

for every compactly supported  $C^1$  tangent vector field  $X$  on  $N$  satisfying

$$\langle X, \nu_{\partial N} \rangle \geq 0$$

at all point of  $\partial N$ , where  $\nu_{\partial N}$  is the unit inward pointing normal of  $\partial N$ . After this short review, we can state one of the main tools of this paper.

**Theorem 2.3** (Strong Maximum Principle for Minimal Hypersurfaces [22]). Suppose that  $N^{m+1}$  is a Riemannian manifold, not necessarily complete, with connected nonempty boundary  $\partial N$  and that  $N^{m+1}$  is mean convex, that is, that

$$\langle \mathbf{H}, \nu_{\partial N} \rangle \geq 0$$

on  $\partial N$ , where  $\mathbf{H}$  is the mean curvature vector of  $\partial N$  and where  $\nu$  is the unit inward pointing normal of  $\partial N$ . Let  $V$  be an  $m$ -dimensional varifold that minimizes area to 1st order in  $N^{m+1}$ . If  $\text{spt} V$  contains point of  $\partial N$ , then it must contain all  $\partial N$ .

### 2.3 Compactness theorems for varifolds

The purpose of this section consists of reviewing the compactness theorems for varifolds that we will need along the paper. These theorems combined with the strong maximum principle for minimal hypersurfaces will be our main tools. Let us start with the definitions of *weak convergence* and *smooth convergence*. Further details can be found in [1, 17, 19].

**Definition 2.2.** Let  $\{V_i\}_{i \in \mathbb{N}}$  be a sequence of varifolds in a smooth manifold  $N$ . We say that  $\{V_i\}$  converges weakly to the varifold  $V$ , if for every smooth function  $\varphi : N \rightarrow \mathbb{R}$  with compact support we have

$$\lim_{i \rightarrow \infty} \int_N \varphi \, d\mu_{V_i} = \int_N \varphi \, d\mu_V,$$

where  $d\mu_{V_i}$  is the Radon measure in  $N$  associated to the varifold  $V_i$ . If the sequence  $\{V_i\}$  converges weakly to  $V$ , then we will write  $V_i \rightharpoonup V$ .

Now let  $M$  be a hypersurface in a Riemannian manifold  $N$ . Given  $p \in M$  and  $r > 0$  we denote by

$$B_r(p) = \{v \in T_p M : |v| < r\}$$

the tangent ball around  $p$  of radius  $r$ . Consider now  $T_p M$  as a vector subspace of  $T_p N$ , and let  $\nu$  be a unit normal vector to  $T_p M$  in  $T_p N$ . Fix a sufficiently small  $\epsilon > 0$  and denote by  $W_{r,\epsilon}(p)$  the solid cylinder around  $p$ , that is,

$$W_{r,\epsilon}(p) := \{ \exp_p(q + t\nu) : q \in B_r(p) \text{ and } |t| < \epsilon \},$$

where  $\exp$  is the exponential map of the Riemannian manifold  $N$ . Given a smooth function  $f : B_r(p) \rightarrow \mathbb{R}$ , the set

$$\text{Graph}(f) := \{ \exp_p(q + f(q)\nu) : q \in B_r(p) \}$$

is called the graph of  $f$  over  $B_r(p)$ .

Now we can define the convergence in  $C^\infty$ -topology.

**Definition 2.3.** Let  $\{M_i\}$  be a sequence of hypersurfaces in a smooth manifold  $N$ . We say that  $\{M_i\}$  converges in  $C^\infty$ -topology with finite multiplicity to a smooth embedded hypersurface  $M$  if

- $M$  consists of accumulations points of  $\{M_i\}$ , that is, for each  $p \in M$  there exists a sequence  $\{p_i\}$  such that  $p_i \in M_i$ , for each  $i \in \mathbb{N}$ , and  $p = \lim_i p_i$ ;
- for every  $p \in M$  there exists  $r, \epsilon > 0$  such that  $M \cap W_{r,\epsilon}(p)$  can be represented as the graph of a function  $f$  over  $B_r(p)$ ;
- for  $i$  large enough, the set  $M_i \cap W_{r,\epsilon}(p)$  consists of a finite number,  $k$ , independent of  $i$ , of graphs of functions  $f_i^1, \dots, f_i^k$  over  $B_r(p)$  that converges smoothly to  $f$ .

The multiplicity of a given point  $p \in M$  is defined by  $k$ . As  $\{M_i\}$  converges smoothly to  $M$ , then we will write  $M_i \rightarrow M$ .

**Remark 2.2.** In general, the limit submanifold is not necessarily connected.

At this point, we can state the weak version of the compactness theorem (see [1, Section 6.4] or [19, Theorem 42.7]).

**Theorem 2.4** (Compactness Theorem for Integral Varifold [1]). Let  $\{M_i\}$  be a sequence of minimal hypersurfaces in  $\mathbb{R}^{n+1}$ , with not necessary the canonical metric, whose area is locally bounded, then a subsequence of  $\{M_i\}$  converges weakly to a stationary integral varifold  $M_\infty$ .

**Remark 2.3.** Notice that in the previous theorem we may have  $M_\infty = \emptyset$ . Indeed, if  $M_i := \{x \in \mathbb{R}^{n+1} : \langle x, e_{n+1} \rangle = i\}$ , then  $\{M_i\}$  is a sequence of minimal hypersurfaces in  $\mathbb{R}^{n+1}$  with the Euclidean metric whose area is locally bounded and  $M_i \rightarrow \emptyset$ .

Actually, when we know that each  $M_i$  is stable, then we can conclude that the convergence above is strong, in the sense that it is smooth away from a closed set, called the *singular set*. This is the content of the next theorem (see [17, Theorem 3] and [25, Theorem 18.1]).

**Theorem 2.5** (Strong Compactness Theorem [17]). Let  $\{M_i\}$  be a sequence of stable minimal hypersurfaces in  $\mathbb{R}^{n+1}$ , with not necessary the canonical metric, with locally bounded area. Then there exist a closed set,  $\text{Sing}$ , and a stationary integral varifolds,  $M_\infty$ , such that a subsequence of  $\{M_i\}$  converges smoothly to  $M_\infty$  away from  $\text{Sing}$ . Moreover, the set  $\text{Sing}$  has Hausdorff dimension at most  $n - 7$ . Hence, it is empty for  $n < 7$ .

As in [12], we will regard all translating solitons as minimal hypersurfaces in  $\mathbb{R}^{n+1}$  with Ilmanen's metric. Then we use the maximum principle and compactness theorems to get a point of contact with a given smooth hypersurface, in our case either hyperplanes parallel to  $\mathbf{e}_{n+1}$  or an element in the family of tilted grim reaper cylinders. In our search for points of contact, it will be interesting to know if a given translator is stable. Hence, we can use the previous theorem to get smooth convergence and to obtain more information about our hypersurface. Similar to what happens to minimal hypersurfaces in  $\mathbb{R}^{n+1}$  with the canonical metric, it is possible to show that graphs are stable. This simple criterion is due to Shahriyari for  $n = 2$ .

**Theorem 2.6** (Shahriyari [18]). Let  $M$  be a translator hypersurface that is a graph over a hyperplane. Then, it is a stable hypersurface in  $\mathbb{R}^{n+1}$  endowed with Ilmanen's metric.

**Remark 2.4.** Although Shahriyari proved the above result for  $n = 2$ , it is straightforward to check that the same proof works for arbitrary dimensions. Xin [31] proved that if  $M$  is a vertical graph, then it is area minimizing as a hypersurface in  $\mathbb{R}^{n+1}$  with Ilmanen's metric.

Another theorem that we will need in this paper is a version of Allard's regularity theorem. This theorem gives us a powerful criterion to decide when the weak convergence is in fact smooth. This version of the Allard's regularity theorem is due to White [30].

**Theorem 2.7** (Allard's Regularity Theorem). Let  $\{M_i\}$  be a sequence of properly embedded minimal hypersurfaces without boundary in  $\mathbb{R}^{n+1}$ , with not necessarily the canonical metric. Suppose that sequence  $\{M_i\}$  in  $\mathbb{R}^{n+1}$  converges weakly to  $S \subset M \subset \mathbb{R}^{n+1}$ , where  $M$  is a connected smoothly embedded hypersurface, and some point in  $M$  has a neighborhood  $U \subset \mathbb{R}^{n+1}$  such that  $\{M_i \cap U\}$  converges weakly to  $M \cap U$  with multiplicity 1, then  $\{M_i\}$  converges to  $M$  smoothly and with multiplicity 1 everywhere.

### 3 The Dynamics Lemma

Fix  $\theta \in [0, \pi/2)$  and define  $u_\theta := -\sin(\theta) \cdot \mathbf{e}_n + \cos(\theta) \cdot \mathbf{e}_{n+1}$ . For a given  $r > 0$ , we consider the cylinder

$$\mathcal{C} := \{x = (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_1^2 + \langle u_\theta, x \rangle^2 \leq r^2\}.$$

Throughout this section  $M^n$  will be a complete, connected, properly embedded translating soliton in  $\mathbb{R}^{n+1}$  such that, outside  $\mathcal{C}$ ,  $M$  is  $C^1$ -asymptotic to two half-hyperplanes  $\mathcal{H}_1$  and  $\mathcal{H}_2$ . In this section, we are going to obtain our main lemma that shows that such a soliton has a surprising amount of internal dynamical periodicity. As a consequence, we shall deduce several properties about the asymptotic geometry of  $M$ .

**Lemma 3.1** (Dynamics Lemma). Let  $M$  be a hypersurface as above. Suppose that  $\{b_i\}_{i \in \mathbb{N}}$  is a sequence in  $[\mathbf{e}_1, u_\theta]^\perp$ , and let  $\{M_i\}_{i \in \mathbb{N}}$  be a sequence of hypersurfaces given by  $M_i := M + b_i$ . Then, after passing to a subsequence,  $\{M_i\}$  converges weakly to a connected stationary integral varifold  $M_\infty$ . Moreover,  $M_\infty$  is smooth outside the cylinder  $\mathcal{C}$  and away from a singular set of Hausdorff dimension at most  $n - 7$ , and the convergence is smooth outside the cylinder and away from the singular set.

**Proof.** The strategy of the proof follows a similar argument as in [12]. However, as we do not assume any hypotheses about the topology (inside the cylinder) besides the fact that we are working in arbitrary dimensions, then we have to use Theorem 2.4 to overcome the difficulties.

From our assumption on  $M$ , there exist  $r > 0$  and two half-hyperplanes,  $\mathcal{H}_1$  and  $\mathcal{H}_2$ , such that the connected components of  $M$  outside  $\mathcal{C}$  are  $C^1$ -asymptotic to them. Let  $w_1$  and  $w_2$  be the unit inward normal vectors of  $\partial\mathcal{H}_1$  and  $\partial\mathcal{H}_2$ , respectively. For every  $\delta > 0$  consider the closed half-hyperplanes  $\mathcal{H}_k(\delta) := \{p + tw_k : p \in \partial\mathcal{H}_k \text{ and } t \geq \delta\}$ ,  $k = 1, 2$ . For  $k = 1, 2$ , let  $\mathcal{Z}_{k,\delta}^+$  denote the half-space in  $\mathbb{R}^{n+1}$  that contains  $\mathcal{H}_{1+k}(\delta)$  and whose boundary contains  $\partial\mathcal{H}_{1+k}(\delta)$  and is perpendicular to  $w_k$ . We also call  $\mathcal{Z}_{k,\delta}^- := (\mathbb{R}^{n+1} - \mathcal{Z}_{k,\delta}^+) \cup \partial\mathcal{Z}_{k,\delta}^+$ , for  $k = 1, 2$ .

Now consider the sets  $M_{k,i}^+(\delta) := M_i \cap \mathcal{Z}_{k,\delta}^+$  and  $M_i^-(\delta) := M_i \cap \mathcal{Z}_{1,\delta}^- \cap \mathcal{Z}_{2,\delta}^-$ , for all  $k \in \{1, 2\}$  (see Figure 5). In order to apply the compactness results in Section 2.3, we must deduce that the sequences  $\{M_{k,i}^+(\delta)\}$  and  $\{M_i^-(\delta)\}$  have locally bounded area. This can be done by using similar ideas as in [12, Lemma 3.1]. Let us explain something about the proof of this fact. First for all, we should show that  $\{M_{k,i}^+(\delta)\}$  has locally bounded area for a sufficiently large  $\delta > 0$ . This can be done by using the fact that  $M_{k,i}^+(\delta)$  can be represented as graph of a function defined over a half-hyperplane. Now we fix a sufficiently large  $\delta$ , using the fact that the boundary of  $M_i^-(\delta)$  can be represented as graph, and obtain that the boundary of the sequence  $\{M_i^-(\delta)\}$  has locally bounded area. Now the area blow-up set defined by

$$\mathcal{U} := \{x \in \mathbb{R}^{n+1} : \limsup \text{Area}(M_i^-(\delta) \cap B(p, r)) = \infty \text{ for every } r > 0\} \quad (3)$$

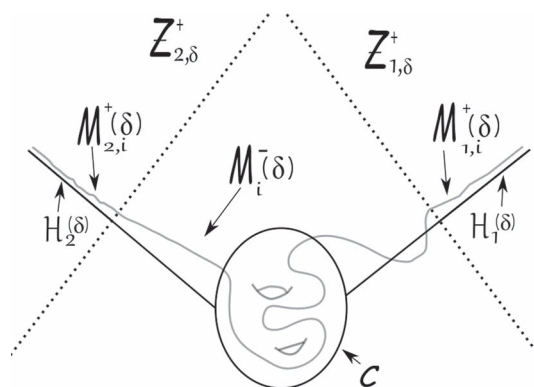


Fig. 5. Transversal section of the behavior of  $M_i$ .

lies inside a large non-horizontal cylinder, where  $B(p, r)$  is the geodesic ball in  $\mathbb{R}^{n+1}$  with Ilmanen's metric. We need to prove that  $\mathcal{U} = \emptyset$  to get that the sequence  $\{M_i^-(\delta)\}$  has locally bounded area too. Arguing by contradiction, let us suppose that  $\mathcal{U} \neq \emptyset$ . In this case, we could take a tilted grim reaper cylinder whose axis is perpendicular to  $\mathcal{C}$  and it does not intersect  $\mathcal{C}$ . Now we could move the tilted grim reaper cylinder until we get a 1st point of contact with  $\mathcal{U}$ , but the strong barrier principle [28, Theorems 2.6 and 7.3] says that  $\mathcal{U}$  must be the tilted grim reaper cylinder, which is absurd.

As the sequence  $\{M_i\}$  has locally bounded area, by Theorem 2.4 there exists a subsequence of  $\{M_i\}$ , which we still denote by  $\{M_i\}$ , that converges weakly to the stationary integral varifold  $M_\infty$ . Furthermore, as outside  $\mathcal{C}$  both connected components of  $M_i$  are graphs (in particular stable by Theorem 2.6), we can apply Theorem 2.5 to conclude that the convergence is smooth outside  $\mathcal{C}$  and away from a singular set with dimension at most  $n - 7$ . This implies that  $M_\infty$  is smooth outside  $\mathcal{C}$  and away from the singular set.

Using this last fact, we can conclude the connectedness of  $M_\infty$  as follows. Taking into account that any loop in  $\mathbb{R}^{n+1}$  intersects  $M_\infty$  in an even number of points (counting multiplicity), then both wings of  $M_\infty$  must lie in the same connected component. Indeed, if this was not true, then we could choose the above-mentioned loop intersecting  $M_\infty$  at one unique point (because  $M_\infty$  is smooth outside a sufficiently large cylinder and away from singular set) that is absurd. This implies that if  $M_\infty$  is not connected, there would be a connected component inside the cylinder. In this case we can consider a suitable tilted grim reaper (whose axis is perpendicular to  $u_\theta$ ) of sufficiently large coordinate in the direction of  $u_\theta$  so that it does not intersect the solid cylinder.

Now if we move it in the direction of  $-u_\theta$  until it touches the component inside the cylinder at 1st point of contact, then we get a contraction because the component inside the cylinder must be the whole tilted grim reaper by Theorem 2.3. Hence,  $M_\infty$  is connected. ■

**Remark 3.1.** Notice that the proof above shows a stronger property. Suppose  $M_i \rightarrow M_\infty$ , where  $\{M_i\}$  and  $M_\infty$  are as in the previous lemma. Take any sequence  $\{c_j\}_{j \in \mathbb{N}}$  in the subspace  $[\mathbf{e}_1, u_\theta]^\perp$ , and define the sequence  $\{\mathfrak{M}_j := M_\infty + c_j\}$  of connected stationary integral varifolds. Reasoning as above, we have that the sequence  $\{\mathfrak{M}_j\}$  has locally bounded area on the compacts sets, and the singular set of each  $\mathfrak{M}_j$  has Hausdorff dimension at most  $n - 7$ . Therefore, by [17, Theorem 3] or [25, Theorem 18.1] we can assume, up to a subsequence, that  $\mathfrak{M}_j \rightarrow \mathfrak{M}_\infty$ , where  $\mathfrak{M}_\infty$  is a stationary integral varifold whose singular set has Hausdorff dimension at most  $n - 7$  and away from the singular it is stable hypersurface. Then we can apply the argument above to conclude that  $\mathfrak{M}_\infty$  satisfies the same properties as  $M_\infty$ .

Now we are going to use the previous lemma to conclude that  $w_1$  and  $w_2$  are parallel to  $u_\theta$ . Moreover, if the half-hyperplanes  $\mathcal{H}_1$  and  $\mathcal{H}_2$  are parts of the same hyperplane, then  $M$  is a hyperplane parallel to  $\mathbf{e}_{n+1}$ .

**Lemma 3.2.** Let  $M$  be a hypersurface as above. Then the half-hyperplanes  $\mathcal{H}_1$  and  $\mathcal{H}_2$  must be parallel to  $\mathbf{e}_{n+1}$ . Moreover, if  $\mathcal{H}_1$  and  $\mathcal{H}_2$  are parts of the same hyperplane  $\Pi$ , then  $M$  must coincide with  $\Pi$ .

**Proof.** We will proceed by contraction. Assume to the contrary that the half-hyperplane

$$\mathcal{H}_1 = \{p + tw_1 : p \in \partial\mathcal{H}_1, t > 0\}$$

is not parallel to direction of translation  $\mathbf{e}_{n+1}$ . Notice that  $\mathbf{e}_j$  and  $E_n$  are perpendicular to  $w_1$  for all  $j \in \{2, \dots, n-1\}$ , where  $E_n := \cos(\theta)\mathbf{e}_n + \sin(\theta)\mathbf{e}_{n+1}$ . Therefore,  $w_1$  is not parallel to  $u_\theta$ . In this case,  $w_1$  forms a nonvanishing angle only with  $\mathbf{e}_1$ , which we denote by  $\alpha := \angle(\mathbf{e}_1, w_1)$ . Suppose that  $\cos \alpha > 0$ . For given real numbers  $t$  and  $l$ , we consider the tilted grim reaper cylinder

$$\mathcal{G}^{t,l} := \{F_\theta(x_1 - t, x_2, \dots, x_n) + t\mathbf{e}_1 + lu_\theta : |x_1 - t| < \frac{\pi}{2\cos(\theta)}, (x_2, \dots, x_n) \in \mathbb{R}^{n-1}\}.$$

By our assumptions about  $M$ , if  $\delta$  is sufficiently large then  $M_1^+(\delta) := M \cap \mathcal{Z}_{1,\delta}^+$  is sufficiently close to  $\mathcal{H}_1$ . From this we can conclude that there exist sufficiently large

$t_0, l_0 \in \mathbb{R}$  so that  $\mathcal{G}^{t_0, l_0}$  do not intersect  $M_1^+(\delta)$  (see Figure 6). In fact, we can choose  $t_0$  so that  $\partial M_1^+(\delta) \cap S_{t_0} = \emptyset$ , where  $S_{t_0} = \left(t_0 - \frac{\pi}{2\cos(\theta)}, t_0 + \frac{\pi}{2\cos(\theta)}\right) \times \mathbb{R}^n$ . Since  $\mathcal{H}_1$  is not parallel to  $\mathbf{e}_{n+1}$  if we translate  $\mathcal{G}^{t_0, l_0}$  at direction of  $-u_\theta$  we conclude that there exists a 1st  $l_1$  such that either  $\mathcal{G}^{t_0, l_0-l_1}$  and  $M_1^+(\delta)$  have a point of contact or  $\text{dist}(\mathcal{G}^{t_0, l_0-l_1}, M_1^+(\delta)) = 0$ .

According to Theorem 2.3 the 1st case cannot be possible because of our assumptions on  $M$ . On the other hand, the 2nd case implies that there exists a sequence  $\{p_i = (p_i^1, \dots, p_i^{n+1})\}$  in  $M_1^+(\delta)$  such that

- the sequence  $\{\langle p_i^{n+1}, u_\theta \rangle\}$  is bounded in  $\mathbb{R}$ ;
- $\lim_i \text{dist}(\mathcal{G}^{t_0, l_0-l_1}, p_i) = 0$ .

Note that the sequence  $\{p_i^1\}$  is bounded (by the asymptotic behavior of  $\mathcal{G}^{t_0, l_0-l_1}$ ). Thus, up to a subsequence, we can suppose  $\{p_i^1\} \rightarrow p_\infty^1$  and  $\{\langle p_i, u_\theta \rangle\} \rightarrow p_\infty^{u_\theta}$ . Let  $\{M_i := M - (0, p_i^2, \dots, p_i^{n+1}) + \langle p_i, u_\theta \rangle u_\theta\}$  be a sequence of hypersurfaces in  $\mathbb{R}^{n+1}$ . By Lemma 3.1, we can suppose that  $M_i \rightarrow M_\infty$ , where  $M_\infty$  is a connected stationary integral varifold. We claim that  $p_\infty = p_\infty^1 \mathbf{e}_1 + p_\infty^{u_\theta} u_\theta \in \text{spt } M_\infty$ . Indeed, take any small open ball  $\mathbb{B}_r(p_\infty)$  of radius  $r$  around  $p_\infty$  in  $\mathbb{R}^{n+1}$ , and let  $p_i^* = p_i^1 \mathbf{e}_1 + \langle p_i, u_\theta \rangle u_\theta$ . Since  $\{p_i^1\} \rightarrow p_\infty^1$ , we obtain that there is an  $i_0$  such that  $\text{dist}_g(p_i^*, p_\infty) < \frac{r}{4}$ . Now fix any  $\epsilon \in (0, \frac{r}{8})$ . Using the definition of limit of varifold (see [1, Section 2.6]), we have

$$\lim_i \mu_{M_i}(\mathbb{B}_r(p_\infty)) = \mu_{M_\infty}(\mathbb{B}_r(p_\infty)),$$

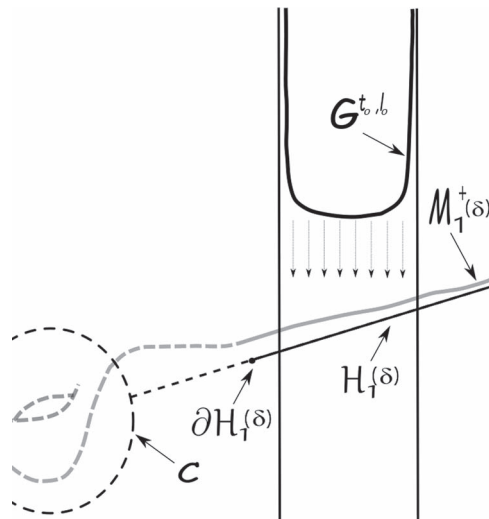


Fig. 6. Transversal section of the behavior of  $M_1^+(\delta)$  and  $\mathcal{G}^{t_0, l_0}$ .

where  $\mu_{M_i}$  is the Radon (Riemannian) measure in  $\mathbb{R}^{n+1}$  associated to  $M_i$ . On the other hand, if  $i > i_0$  we have  $\mu_{M_i}(\mathbb{B}_r(p_\infty)) \geq \mu_{M_i}(\mathbb{B}_\epsilon(p_i^*)) \geq c(n, \epsilon)$  by the monotonicity formula for varifold, where  $c(n, \epsilon)$  is a positive constant that depends only on  $n$  and  $\epsilon$ , because each  $M_i$  is a smooth hypersurface. In particular, we have

$$\mu_{M_\infty}(\mathbb{B}_r(p_\infty)) \geq c(n, \epsilon) > 0,$$

that is,  $p_\infty \in \text{spt } M_\infty$ , and follows that  $\text{spt } M_\infty$  and  $\mathcal{G}^{t_0, l_0-l_1}$  have a point of contact at  $p_\infty$ . Therefore, by Theorem 2.3 we must have  $M_\infty = \mathcal{G}^{t_0, l_0-l_1}$ . But this is clearly impossible by our assumption about  $w_1$  and Theorem 2.7. Analogously, we can conclude that  $\cos \alpha$  cannot be negative and that  $\mathcal{H}_2$  is parallel to  $\mathbf{e}_{n+1}$ .

Finally, if  $\mathcal{H}_1$  and  $\mathcal{H}_2$  are part of the same hyperplane  $\Pi$ , which we suppose to be  $[\mathbf{e}_1]^\perp$ , then we claim that the 1st coordinate must be constant on  $M$ . In fact, suppose to the contrary that this is true. In this case, the 1st coordinate  $x_1$  takes an extreme value either at point in  $M$  or along of a sequence  $\{p_i = (p_i^1, \dots, p_i^{n+1})\}$  such that  $\{\langle p_i, u_\theta \rangle\} \rightarrow p_\infty^{u_\theta}$ . From Theorem 2.3 the 1st case is impossible. Regarding the 2nd case, suppose that  $\{x_1(p_i)\} \rightarrow \sup_M x_1 (> 0)$ , and let us denote  $\Pi_1 := \sup_M x_1 \mathbf{e}_1 + \text{span}[\mathbf{e}_2, \dots, \mathbf{e}_n, u_\theta]$ . Again, if we consider the sequence

$$\{M_i := M - (0, p_i^2, \dots, p_i^{n+1}) + \langle p_i, u_\theta \rangle u_\theta\}$$

by Lemma 3.1, a subsequence converges to  $M_\infty$ , where  $M_\infty$  is a connected stationary integral varifold, thus (reasoning as above) we have an interior point of contact between  $\text{spt } M_\infty$  and  $\Pi_1$ . So, by Theorem 2.3 we conclude that  $M_\infty = \Pi_1$ , which is impossible. This shows that the 1st coordinate  $x_1$  is constant. Therefore,  $M$  must be the hyperplane  $\Pi$ . ■

Let us finish this section with another application of Lemma 3.1.

**Lemma 3.3.** Let  $M$  be a hypersurface satisfying the previous conditions and assume that the half-hyperplanes  $\mathcal{H}_1$  and  $\mathcal{H}_2$  are distinct. Consider a domain  $\Sigma$  of  $M$  (not necessarily compact) with nonempty boundary  $\partial\Sigma$  such that the function  $x \mapsto \langle x, u_\theta \rangle$  of  $\Sigma$  is bounded. Then the supremum and the infimum of the  $x_1$ -coordinate function of  $\Sigma$  are reached along the boundary of  $\Sigma$ , that is, there exists no sequence  $\{p_i\}$  in the interior of  $\Sigma$  such that  $\lim_{i \rightarrow \infty} \text{dist}(p_i, \partial\Sigma) > 0$  and either  $\lim_{i \rightarrow \infty} x_1(p_i) = \sup_\Sigma x_1$  or  $\lim_{i \rightarrow \infty} x_1(p_i) = \inf_\Sigma x_1$ .

**Proof.** The proof of this lemma follows the same ideas as in [12]. The only difference is that in the proof we should use the Lemma 3.1 and Theorem 2.3 to conclude. ■

## 4 The Proof of the Main Theorems

### 4.1 The case $\theta \in [0, \pi/2)$

This section is devoted to demonstrating the main theorem for  $\theta \in [0, \pi/2)$ . As in the previous section, we fix  $\theta \in [0, \pi/2)$  and define  $u_\theta = -\sin(\theta) \cdot \mathbf{e}_n + \cos(\theta) \cdot \mathbf{e}_{n+1}$ .

**Theorem 4.1.** Let  $f : M \rightarrow \mathbb{R}^{n+1}$  be a complete, connected, properly embedded translating soliton, and consider  $\mathcal{C} := \{x \in \mathbb{R}^{n+1} : \langle x, \mathbf{e}_1 \rangle^2 + \langle u_\theta, x \rangle^2 \leq r^2\}$ , where  $r > 0$ . Assume that  $M$  is  $C^1$ -asymptotic to two half-hyperplanes outside  $\mathcal{C}$ . Then we have one, and only one, of these two possibilities:

- (a) Both half-hyperplanes are contained in the same hyperplane  $\Pi$  parallel to  $\mathbf{e}_{n+1}$ , and  $M$  coincides with  $\Pi$ ;
- (b) Both half-hyperplanes are included in different parallel hyperplanes, and  $M$  coincides with a tilted grim reaper cylinder.

We will divide the proof of the theorem in many lemmas. But, before we start with the proof, we have to introduce some notation that we will use throughout the whole section.

Following [11] and [12], let us define the foliation of  $\mathbb{R}^{n+1}$  given by

$$\Pi(t) = \{x \in \mathbb{R}^{n+1} : \langle x, \mathbf{e}_1 \rangle = t\}. \quad (4)$$

Furthermore, given  $A \subset \mathbb{R}^{n+1}$  and  $t \in \mathbb{R}$ , let us define the sets

$$A_+(t) = \{x \in A : \langle x, \mathbf{e}_1 \rangle \geq t\}, \quad A_-(t) = \{x \in A : \langle x, \mathbf{e}_1 \rangle \leq t\}$$

$$A^+(t) = \{x \in A : \langle x, u_\theta \rangle \geq t\}, \quad A^-(t) = \{x \in A : \langle x, u_\theta \rangle \leq t\}.$$

Recall that we are assuming that the translating velocity is  $\mathbf{e}_{n+1}$ . From Lemma 3.2, we work only in the case when the half-hyperplanes  $\mathcal{H}_1$  and  $\mathcal{H}_2$  lie in different and parallel hyperplanes to  $\mathbf{e}_{n+1}$ . So, up to a translation, we can assume that the half-hyperplanes are contained in  $\Pi(-\delta)$  and  $\Pi(\delta)$ , for a certain  $\delta > 0$ . Let us begin by proving that both half-hyperplanes are parallel to  $u_\theta$ .

**Lemma 4.1.** The two connected components of  $M$  that lie outside the cylinder  $\mathcal{C}$  point in the same direction of  $u_\theta$ .

**Proof.** First of all, notice that  $M$  cannot be asymptotic to the half-hyperplanes

$$\mathcal{H}_1 = \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle < r_1 < 0, x_1 = -\delta\}$$

and

$$\mathcal{H}_2 = \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle < r_2 < 0, x_1 = \delta\}.$$

This is a trivial consequence of Theorem 2.3, when one compares  $M$  with a suitable copy of a tilted grim reaper transverse to the hyperplane  $\Pi(0)$  (as we did at the end of the proof of Lemma 3.1).

For the remaining cases, we proceed again by contradiction. Suppose at first that

$$\mathcal{H}_1 = \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle > r_1 > 0, x_1 = -\delta\}$$

and

$$\mathcal{H}_2 = \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle < r_2 < 0, x_1 = \delta\}$$

for some  $r_1 > 0$  and  $r_2 < 0$ . Given  $t$  and  $l$  in  $\mathbb{R}$ , let  $\mathcal{G}^{t,l}$  be the tilted grim reaper cylinder defined by

$$\mathcal{G}^{t,l} := \left\{ F_\theta(x_1 - t, x_2, \dots, x_n) + t\mathbf{e}_1 + lu_\theta : |x_1 - t| < \frac{\pi}{2\cos(\theta)}, (x_2, \dots, x_n) \in \mathbb{R}^{n-1} \right\}. \quad (5)$$

Consider  $\mathcal{G}^{\frac{\pi}{2\cos(\theta)} + \delta, 0}$ , which lies in  $(\delta, \delta + \frac{\pi}{\cos(\theta)}) \times \mathbb{R}^n$  (see Figure 7). Note that it is asymptotic to the half-hyperplanes  $\Pi(\delta)$  and  $\Pi(\delta + \frac{\pi}{\cos(\theta)})$ . Fix  $\epsilon \in (0, 2\delta)$ . Using the fact that  $\mathcal{G}^{\frac{\pi}{2\cos(\theta)} + \delta, 0}$  is asymptotic to the half-hyperplanes outside the cylinder, then there exists  $\delta_1 > r_1$ , depending only on  $\epsilon$ , such that (Here we are using the same notation of Lemma 3.1.)

$$\mathcal{G}^{\frac{\pi}{2\cos(\theta)} + \delta, 0} \cap \mathcal{Z}_{\delta_1}^+ \subset [(\delta, \delta + \frac{\epsilon}{2}) \times \mathbb{R}^n] \cap \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle > \delta_1\}. \quad (6)$$

On the other hand, taking into account the asymptotic behavior of  $M$  and our assumptions about the wings, there exists a  $\delta_2 > -r_2$ , depending only on  $\epsilon$ , such that

$$M \cap \mathcal{Z}_{\delta_2}^+ \subset [(\delta - \frac{\epsilon}{2}, \delta + \frac{\epsilon}{2}) \times \mathbb{R}^n] \cap \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle < \delta_2\}. \quad (7)$$

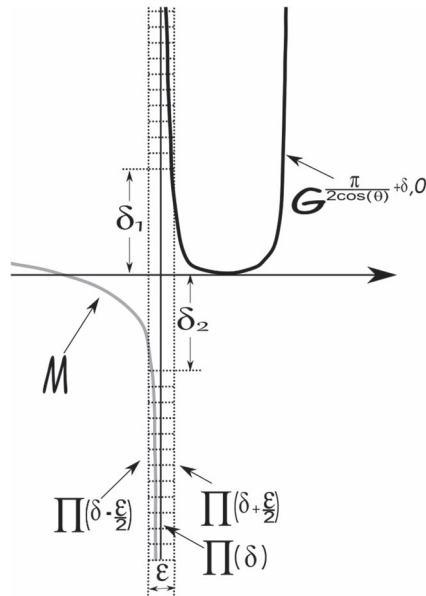


Fig. 7. Transversal section of the  $M_1^+(\delta)$  and  $\mathcal{G}^{\frac{\pi}{2\cos(\theta)}+\delta,0}$ .

From (6) and (7), there exists a  $t > 0$  such that the tilted grim reaper cylinder  $\mathcal{G}^{\frac{\pi}{2\cos(\theta)}+\delta+t, -\delta_1-\delta_2-1}$  satisfies

$$\mathcal{G}^{\frac{\pi}{2\cos(\theta)}+\delta+t, -\delta_1-\delta_2-1} \cap M = \emptyset.$$

Now since  $\epsilon \in (0, 2\delta)$ , there is a finite  $t_0$  such that either  $M$  and  $\mathcal{G}^{\frac{\pi}{2\cos(\theta)}+\delta+t_0, -\delta_1-\delta_2-1}$  have a 1st point of contact or there is a sequence  $\{p_i = (p_i^1, \dots, p_i^{n+1})\}$  in  $M$  satisfying the following conditions:

- i.  $\{\langle p_i, u_\theta \rangle\}$  is a bounded sequence;
- ii.  $\{(0, p_i^2, \dots, p_i^{n+1}) - \langle p_i, u_\theta \rangle u_\theta\}$  is an unbounded sequence;
- iii.

$$\lim_i \left\{ \text{dist} \left( p_i, \mathcal{G}^{\frac{\pi}{2\cos(\theta)}+\delta+t_0, -\delta_1-\delta_2-1} \right) \right\} = 0. \quad (8)$$

Notice that in this last case the sequence  $\{p_i^1\}$  is bounded because of the asymptotic behavior of  $M$ . Thus, we can suppose  $\{p_i^1\} \rightarrow p_\infty^1$  and  $\{\langle p_i, u_\theta \rangle\} \rightarrow p_\infty^{u_\theta}$ . In particular, from (8), we have

$$p_\infty^1 \mathbf{e}_1 + p_\infty^{u_\theta} u_\theta \in \mathcal{G}^{\frac{\pi}{2\cos(\theta)}+\delta+t_0, -\delta_1-\delta_2-1}.$$

According to Theorem 2.3 and the asymptotic behavior of  $M$  the 1st case cannot happen. Regarding the 2nd case, let us define the sequence

$$\{M_i := M - (0, p_i^2, \dots, p_i^{n+1}) + \langle p_i, u_\theta \rangle u_\theta\}.$$

By Lemma 3.1, up to a subsequence, we have that  $M_i \rightharpoonup M_\infty$ , where  $M_\infty$  is a connected stationary integral varifold. Arguing as in Lemma 3.2, we have

$$p_\infty^1 \mathbf{e}_1 + p_\infty^{u_\theta} u_\theta \in \text{spt } M_\infty \cap \mathcal{G}^{\frac{\pi}{2 \cos(\theta)} + \delta + t, -\delta_1 - \delta_2 - 1}.$$

Thus, by Theorem 2.3 we get

$$M_\infty = \mathcal{G}^{\frac{\pi}{2 \cos(\theta)} + \delta + t_0, -\delta_1 - \delta_2 - 1}.$$

But this is impossible by the asymptotic behavior of  $M$ .

The case when

$$\mathcal{H}_1 = \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle < r_1 < 0, x_1 = -\delta\}$$

and

$$\mathcal{H}_2 = \{x \in \mathbb{R}^{n+1} : \langle x, u_\theta \rangle > r_2 > 0, x_1 = \delta\}$$

can be excluded using a symmetric argument. This concludes the proof.  $\blacksquare$

We shall see now that our embedded soliton must lie in the slab limited by the hyperplanes  $\Pi(-\delta)$  and  $\Pi(\delta)$ .

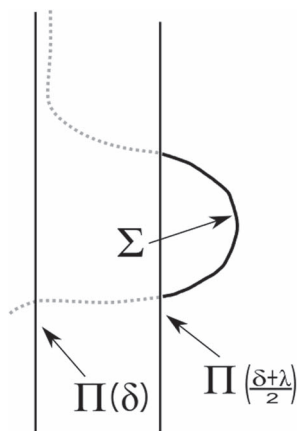
**Lemma 4.2.**  $M$  lies inside the slab  $S := (-\delta, \delta) \times \mathbb{R}^n$ .

**Proof.** Assume that  $\lambda := \sup_M x_1 > \delta$  and consider the domain

$$\Sigma = \{x \in M : \langle x, \mathbf{e}_1 \rangle \geq \frac{\delta + \lambda}{2}\}.$$

The asymptotic behavior of  $M$  tells us that the function  $x \mapsto \langle x, u_\theta \rangle$  is bounded in  $\Sigma$  (see Figure 8). Now by Lemma 3.3 we have

$$\lambda = \sup_\Sigma x_1 = \sup_{\partial \Sigma} x_1. \quad (9)$$



**Fig. 8.** Transversal section of the  $\Sigma$ .

Therefore, because  $\partial\Sigma \subset \Pi\left(\frac{\delta+\lambda}{2}\right)$  it follows that  $x_1(p) = \frac{\delta+\lambda}{2}$  for all  $p \in \Sigma$ , which is absurd. Moreover, if  $x_1(p) = \delta$  for any  $p \in M$ , then  $M$  and the hyperplane  $\Pi(\delta)$  have a point of contact. Hence, by Theorem 2.3 we must have  $M = \Pi(\delta)$  and again we arrive at a contradiction. This finishes the proof that  $x_1(p) < \delta$  for all  $p \in M$ . Using the same idea we shall obtain that  $x_1(p) > -\delta$  for all  $p \in M$ . ■

Now we are going to show that the distance between the two half-hyperplanes is precisely  $\frac{\pi}{\cos(\theta)}$ , like in the tilted grim reaper cylinder.

**Lemma 4.3.** We have  $2\delta = \frac{\pi}{\cos(\theta)}$ .

**Proof.** We argue again by contradiction. Assume at first that  $2\delta > \frac{\pi}{\cos(\theta)}$ . By the asymptotic behavior of  $M$  we can place a tilted grim reaper cylinder  $\mathcal{G}^{0,l}$  inside  $S$ , for sufficiently large  $l$ , so that  $\mathcal{G}^{0,l} \cap M = \emptyset$  (see Figure 9).

Now consider  $\mathcal{A} := \{l \in \mathbb{R} : \mathcal{G}^{0,l} \cap M = \emptyset\}$ , and let  $l_0 = \inf \mathcal{A}$ . Note that  $l_0 > -\infty$  by the asymptotic behavior of  $M$ . If  $l_0 \notin \mathcal{A}$ , then  $M$  and  $\mathcal{G}^{0,l_0}$  have a point of contact. So  $M = \mathcal{G}^{0,l_0}$  by Theorem 2.3. But this is impossible. On the other hand, if  $l_0 \in \mathcal{A}$  then  $\text{dist}(M, \mathcal{G}^{0,l_0}) = 0$ . This means that there exists a sequence  $\{p_i = (p_i^1, \dots, p_i^{n+1})\}$  in  $M$  such that the sequences  $\{p_i^1\}$  and  $\{p_i, u_\theta\}$  are bounded, the sequence  $\{(0, p_i^2, \dots, p_i^{n+1}) - \langle p_i, u_\theta \rangle u_\theta\}$  is unbounded and  $\text{dist}(p_i, \mathcal{G}^{0,l_0}) = 0$ . Thus, up to a subsequence, one has  $p_i^1 \rightarrow p_\infty^1$  and  $\langle p_i, u_\theta \rangle \rightarrow p_\infty^{u_\theta}$ . Then we consider the sequence of hypersurfaces  $\{M_i\}$ ,

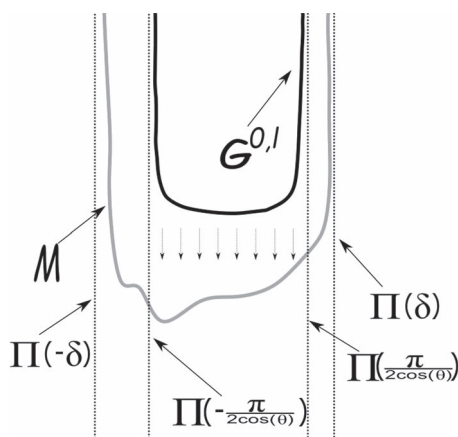


Fig. 9. Transversal section of the behavior of  $\mathcal{G}^{0,l}$  with respect to  $M$ .

where

$$M_i := M - (0, p_i^2, \dots, p_i^{n+1}) + \langle p_i, u_\theta \rangle u_\theta.$$

Using Lemma 3.1 we can suppose that  $M_i \rightarrow M_\infty$ , where  $M_\infty$  is a connected stationary integral varifold with  $p_\infty^1 \mathbf{e}_1 + p_\infty^{u_\theta} u_\theta \in \text{spt } M_\infty$ . Hence,  $p_\infty^1 \mathbf{e}_1 + p_\infty^{u_\theta} u_\theta$  is a point of contact between  $\text{spt } M_\infty$  and  $\mathcal{G}^{0,l_0}$ . Thus, again by Theorem 2.3 we get that  $\mathcal{G}^{0,l_0} = M$ , which contradicts our assumptions about the behavior of  $M$ . Consequently,  $2\delta \leq \frac{\pi}{\cos(\theta)}$ . Comparing  $M$  with a tilted grim reaper cylinder “outside”  $M$  we conclude  $2\delta = \frac{\pi}{\cos(\theta)}$ . This completes the proof. ■

In the next lemma we prove that the connected components of  $M \setminus \mathcal{C}$ , which we will call from now on *the wings of  $M$* , are graphs.

**Lemma 4.4.** If  $t > 0$  is sufficiently large, then the two connected components of  $M^+(t)$  are graphs over the hyperplane  $[\mathbf{e}_{n+1}]^\perp$ .

**Proof.** Observe that, if we take a sufficiently large  $t$ , then

$$M^+(t) \subset M_+ \left( \frac{\pi}{2 \cos(\theta)} - \tau \right) \cup M_- \left( -\frac{\pi}{2 \cos(\theta)} + \tau \right),$$

for a small enough  $\tau > 0$ . Therefore, we only need to prove that if  $\delta$  is small enough, then  $M_+ \left( \frac{\pi}{2 \cos(\theta)} - \tau \right)$  is a graph over a subset of  $[\mathbf{e}_{n+1}]^\perp$ . The case of  $M_- \left( -\frac{\pi}{2 \cos(\theta)} + \tau \right)$  is treated in a similar way.

Fix a sufficiently small  $\epsilon > 0$ , with  $\epsilon < \frac{1}{8}$ . Since  $\mathcal{G}(= \mathcal{G}^{0,0})$  and  $M \setminus \mathcal{C}$  are  $C^1$ -asymptotic to the same half-hyperplane contained in  $\Pi\left(\frac{\pi}{2\cos(\theta)}\right)$ , we can represent  $M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)$  as graph over  $\mathcal{G}$ . Hence, we can find a smooth map  $\varphi : T_\tau := \left(\frac{\pi}{2\cos(\theta)} - \tau, \frac{\pi}{2\cos(\theta)}\right) \longrightarrow \mathbb{R}$  such that

$$\sup_{T_\tau} |\varphi| < \epsilon \text{ and } \sup_{T_\tau} |D\varphi| < \epsilon. \quad (10)$$

Moreover, the map  $\tilde{F} : T_\tau \times \mathbb{R}^{n-1} \longrightarrow \mathbb{R}^{n+1}$  given by

$$\tilde{F} = F_\theta + \varphi v_F \quad (11)$$

is a parametrization of  $M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)$ , where  $F_\theta$  is the parametrization given by (2) and

$$v_{F_\theta}(x_1, \dots, x_n) = \sin(x_1 \cos(\theta))\mathbf{e}_1 - \cos(x_1 \cos(\theta))u_\theta.$$

Let us define the map  $\Pi : \mathbb{R}^{n+1} \longrightarrow \mathbb{R}^n$  given by  $\Pi(x_1, \dots, x_{n+1}) = (x_1, \dots, x_n)$  and consider its restriction

$$\tilde{\Pi} := \Pi|_{\text{int}\left(M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)\right)} : \text{int}\left(M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)\right) \longrightarrow T_\tau. \quad (12)$$

Note that the image of  $\tilde{\Pi}$  lies on  $T_\tau$ , because for all  $x \in \text{int}\left(M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)\right)$  we have

$$\frac{\pi}{2\cos(\theta)} - \tau < \langle x, \mathbf{e}_1 \rangle < \frac{\pi}{2\cos(\theta)},$$

by the definition of  $M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)$ . The idea here consists of showing that  $\tilde{\Pi}$  is a diffeomorphism. To deduce this, by a standard topological argument, we only must check that

1.  $\tilde{\Pi}$  is a proper covering map;
2.  $\text{int}\left(M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)\right)$  is path connected.

First, let us show that  $\tilde{\Pi}$  is a local diffeomorphism. Equivalently, let us show that  $H > 0$  on  $M_+\left(\frac{\pi}{2\cos(\theta)} - \tau\right)$ . Here we will use the parametrization (11). To do this, we use the orthonormal frame  $\{E_i\}$  given by

$$\{E_1 := \cos(x_1 \cos(\theta))\mathbf{e}_1 + \sin(x_1 \cos(\theta))u_\theta, E_j := \mathbf{e}_j, j \in \{2, \dots, n-1\} \quad (13)$$

$$E_n := \cos(\theta)\mathbf{e}_n + \sin(\theta)\mathbf{e}_{n+1}, v_F\}.$$

Representing the vectors with respect to this basis, we obtain that the unit normal  $N_{\tilde{F}}$  of  $\tilde{F}$  is given by the formula

$$\Omega \cdot N_{\tilde{F}} = AE_1 + B \sum_{j=2}^{n-1} (-1)^j \left[ n - \frac{j+1}{2} \right] \partial_{x_j} \varphi E_j + C \partial_{x_n} \varphi E_n + B v_F, \quad (14)$$

where

$$A := (-1)^{n-2} (\sin(\theta) \sin(x_1 \cos(\theta)) \partial_{x_n} \varphi - \cos(x_1 \cos(\theta)) \partial_{x_1} \varphi) \quad (15)$$

$$B := 1 + \varphi \cos(\theta) \cos(x_1 \cos(\theta)) \quad (16)$$

$$C := (-1)^{n-1} \cos(\theta) (1 + \varphi \cos(\theta) \cos(x_1 \cos(\theta))) \quad (17)$$

$$\begin{aligned} \Omega^2 &:= [\sin(\theta) \sin(x_1 \cos(\theta)) \partial_{x_n} \varphi - \cos(x_1 \cos(\theta)) \partial_{x_1} \varphi]^2 \\ &+ [1 + \varphi \cos(\theta) \cos(x_1 \cos(\theta))]^2 \left( 1 + \sum_{j=2}^n (\partial_{x_j} \varphi)^2 \right) \\ &+ \cos^2(\theta) [1 + \varphi \cos(\theta) \cos(x_1 \cos(\theta))]^2 (\partial_{x_n} \varphi)^2. \end{aligned} \quad (18)$$

In particular by (1), we have

$$\begin{aligned} \frac{\Omega \cdot H}{\cos \theta \cos(x_1 \cos(\theta))} &:= 1 + \varphi \cos(\theta) \cos(x_1 \cos(\theta)) + (-1)^n \sin(x_1 \cos(\theta)) \partial_{x_1} \varphi \\ &+ (-1)^n \sin(\theta) \cos(x_1 \cos(\theta)) \partial_{x_n} \varphi + (-1)^n \cos(\theta) \sin(\theta) \varphi \partial_{x_n} \varphi. \end{aligned} \quad (19)$$

Now by our assumptions about  $\epsilon$ ,  $\varphi$ , and  $D\varphi$  we immediately deduce that  $H > 0$  at all  $p \in M_+ \left( \frac{\pi}{2 \cos(\theta)} - \tau \right)$ . Hence,  $\tilde{\Pi}$  is a local diffeomorphism.

The previous argument also implies that  $\tilde{\Pi}$  is onto. Indeed, if it does not there would be a vertical cylinder that intersects  $T_\tau$  but it would not intersect the set  $M_+ \left( \frac{\pi}{2 \cos(\theta)} - \tau \right)$ . Taking into account the asymptotic behavior of  $M$ , we could translate horizontally this cylinder until having a 1st contact with

$$\text{int} \left( M_+ \left( \frac{\pi}{2 \cos(\theta)} - \tau \right) \right).$$

At this 1st contact the normal vector field to  $M$  would be horizontal, which is absurd because we have proved that  $H > 0$  on  $M_+ \left( \frac{\pi}{2 \cos(\theta)} - \tau \right)$ .

Finally, let us check that  $\tilde{\Pi}$  is proper. Let  $K \subset T_\tau$  a compact set and  $\{p_i\}_{i \in \mathbb{N}}$  be a sequence on  $\tilde{\Pi}^{-1}(K)$ . Note that the sequence  $\{p_i\}_{i \in \mathbb{N}}$  is bounded, because of the asymptotic behavior of  $M$  and the fact that  $\text{dist}(K, \partial T_\tau) > 0$ . So, up to a subsequence, we can assume that  $p_i \rightarrow p_\infty$ . Since the set  $\tilde{\Pi}^{-1}(K)$  is closed, it follows that  $p_\infty \in \tilde{\Pi}^{-1}(K)$ . This proves that  $\tilde{\Pi}^{-1}(K)$  is compact.

At this point, we have that any connected component of  $\text{int}\left(M_+ \left(\frac{\pi}{2\cos(\theta)} - \tau\right)\right)$  is a graph over  $T_\tau$ . But only one of them contains the wing. This means that if there were another connected component,  $\Sigma$ , then the function  $x \mapsto \langle x, u_\theta \rangle$  would be bounded on  $\Sigma$  and  $\partial\Sigma \subset \Pi\left(\frac{\pi}{2\cos(\theta)} - \tau\right)$ , which is contrary to Lemma 3.3. Repeating the same argument we should obtain that  $M_- \left(-\frac{\pi}{2\cos(\theta)} + \tau\right)$  is a graph over the hyperplane  $[\mathbf{e}_{n+1}]^\perp$ . ■

Now we are going to show that it is possible to place a tilted grim reaper cylinder below  $M$ . This means that  $M$  lies in the convex region limited by the tilted grim reaper cylinder. Without loss of generality we can assume from now on that  $\inf_M \langle x, u_\theta \rangle = 0$ .

**Lemma 4.5.** There is a tilted grim reaper cylinder that contains  $M$  “inside” it, that is,  $M$  lies in the convex region of the complement of a tilted grim reaper cylinder.

**Proof.** Consider the family of “half”-tilted grim reaper cylinders

$$\mathcal{G}_\pm^{t,-\epsilon} := \{x \in \mathcal{G}^{0,-\epsilon} : \pm \langle x, \mathbf{e}_1 \rangle \geq 0\} \pm t\mathbf{e}_1, \quad (20)$$

where  $\epsilon > 0$  is fixed and  $t \in [0, \infty)$ .

Let us work with the “half”-tilted grim reaper cylinder  $\mathcal{G}_+^{t,-\epsilon}$ . By taking a sufficiently large  $t_0$ , we obtain  $\mathcal{G}_+^{t_0,-\epsilon} \cap M = \emptyset$ . Hence, the set  $\mathcal{A}$  defined by

$$\mathcal{A} := \{t \in [0, \infty) : \mathcal{G}_+^{t,-\epsilon} \cap M = \emptyset\}$$

is not empty. Take  $s_0 = \inf \mathcal{A}$ . We claim that  $s_0 = 0$ . Otherwise, we have two possibilities for  $s_0 > 0$ : either  $s_0 \in \mathcal{A}$  or  $s_0 \notin \mathcal{A}$ . If  $s_0 \notin \mathcal{A}$  then  $\mathcal{G}_+^{s_0,-\epsilon} \cap M \neq \emptyset$  and since  $\partial\mathcal{G}_+^{s_0,-\epsilon} \cap M = \emptyset$ , we conclude that  $\mathcal{G}_+^{s_0,-\epsilon} \subset M$ , by Theorem 2.3, but this is absurd because

$$0 = \inf_M \langle x, u_\theta \rangle > \inf_{\mathcal{G}_+^{s_0,-\epsilon}} \langle x, u_\theta \rangle = -\epsilon.$$

If  $s_0 \in \mathcal{A}$  then  $\text{dist}(\mathcal{G}_+^{s_0,-\epsilon}, M) = 0$ . This tells us that there exists a sequence  $\{p_i = (p_i^1, \dots, p_i^{n+1})\}$  in  $M$  such that

- i.  $\lim_i \text{dist}(p_i, \mathcal{G}_+^{s_0,-\epsilon}) = 0$ ;
- ii.  $\{p_i^1\} \rightarrow p_\infty^1$  and  $a < p_i^1 - t < b$ , where  $0 < a < b < \frac{\pi}{2\cos(\theta)}$  are constants;
- iii.  $\{\langle p_i, u_\theta \rangle\} \rightarrow p_\infty^{u_\theta}$ ;
- iv. the sequence  $\{(0, p_i^2, \dots, p_i^{n+1}) - \langle p_i, u_\theta \rangle u_\theta\}$  is unbounded.

In this case, consider the sequence of hypersurfaces

$$\{M_i = M - (0, p_i^2, \dots, p_i^{n+1}) + \langle p_i, u_\theta \rangle u_\theta\}.$$

By Lemma 3.1 we can suppose  $M_i \rightarrow M_\infty$ , where  $M_\infty$  is a connected stationary integral varifold. Since  $p_\infty^1 \mathbf{e}_1 + p_\infty^\theta u_\theta \in \mathcal{G}_+^{t, -\epsilon} \cap \text{spt } M_\infty$ , using again Theorem 2.3 we get  $\mathcal{G}_+^{t, -\epsilon} = M_\infty$ . But this again contradicts the asymptotic behavior of  $M$ . Therefore,  $\inf \mathcal{A} = 0$ , and

$$\mathcal{G}_+^{0, -\epsilon} \cap M = \emptyset,$$

for all  $\epsilon > 0$ . A symmetric argument shows that  $\mathcal{G}_-^{0, -\epsilon} \cap M = \emptyset$ . Thus,  $\mathcal{G}^{0, -\epsilon} \cap M = \emptyset$ .

This completes the proof.  $\blacksquare$

As an application of the previous lemma, we will conclude that our hypersurface is in fact a graph over the hyperplane  $[\mathbf{e}_{n+1}]^\perp$ .

**Lemma 4.6.**  $M$  is a graph over  $(-\frac{\pi}{2\cos\theta}, \frac{\pi}{2\cos\theta}) \times \mathbb{R}$ .

**Proof.** For each  $i \in \mathbb{N}$  consider the sets

$$T_i := \{v \in \mathbb{R}^{n+1} : \langle v, E_n \rangle \geq i\},$$

where  $E_n := \cos(\theta)\mathbf{e}_n + \sin(\theta)\mathbf{e}_{n+1}$ , and call  $\alpha := \lim_i \inf_{T_i \cap M} \langle x, u_\theta \rangle$ . Consider a sequence  $\{p_i = (p_i^1, \dots, p_i^{n+1})\}$  in  $M$  such that

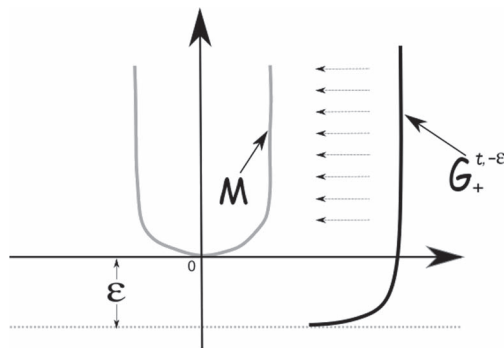
- i.  $p_i \in T_i \cap M$  and  $\langle p_i, u_\theta \rangle - \inf_{T_i \cap M} \langle x, u_\theta \rangle < \frac{1}{i}$ ;
- ii.  $\{p_i^1\} \rightarrow p_\infty^1$  and  $-\frac{\pi}{2\cos(\theta)} < p_\infty^1 < \frac{\pi}{2\cos(\theta)}$ ;
- iii.  $\{\langle p_i, u_\theta \rangle\} \rightarrow \alpha$ .

Consider again the sequence of hypersurfaces

$$\{M_i = M - (0, p_i^2, \dots, p_i^{n+1}) + \langle p_i, u_\theta \rangle u_\theta\}.$$

By Lemma 3.1, we can suppose  $M_i \rightarrow M_\infty$ , where  $M_\infty$  is a connected stationary integral varifold. Since  $p_\infty^1 \mathbf{e}_1 + \alpha u_\theta \in \text{spt } M_\infty$ , it follows that  $\inf_{\text{spt } M_\infty} \langle x, u_\theta \rangle \leq \alpha$ . We claim that  $\alpha = \inf_{\text{spt } M_\infty} \langle x, u_\theta \rangle$ . Indeed, take any  $p \in \mathbb{R}^{n+1}$  such that  $\langle p, u_\theta \rangle < \alpha$ . Let  $B_r(p)$  be the open ball in  $\mathbb{R}^{n+1}$ , where  $r \in (0, \frac{\alpha - \langle p, u_\theta \rangle}{4})$ . Note that  $B_r(p) \cap \Pi_\alpha = \emptyset$ , where  $\Pi_\alpha = [u_\theta]^\perp + \alpha u_\theta$ . Take any  $\epsilon \in (0, \frac{\alpha - \langle p, u_\theta \rangle}{4})$ . By the definition of  $\alpha$ , there is an  $i_0$  such that if  $i > i_0$  then

$$\inf_{T_i \cap M} \langle x, u_\theta \rangle > \alpha - \epsilon > 0. \quad (21)$$



**Fig. 10.** Transversal section of the behavior of  $\mathcal{G}_+^{t, -\epsilon}$  with respect to  $M$ .

Let  $\varphi : \mathbb{R}^{n+1} \rightarrow [0, \infty)$  be a smooth function such that

- a.  $\text{spt } \varphi \subset B_r(p)$ ;
- b.  $\varphi|_{B_{\frac{r}{2}}(p)} \equiv 1$ .

By (21), if  $i > i_0$  we have

$$\int_{\mathbb{R}^{n+1}} \varphi \, d\mu_{M_i} = 0,$$

where  $d\mu_{M_i}$  denotes the Radon (Riemannian) measure associated to  $M_i$ . This implies that  $\int_{\mathbb{R}^{n+1}} \varphi \, d\mu_{M_\infty} = 0$  and so  $p \notin \text{spt } M_\infty$ . Consequently,  $\alpha = \inf_{\text{spt } M_\infty} \langle x, u_\theta \rangle$ . Now we claim that  $M_\infty$  coincides with the tilted grim reaper cylinder. The proof follows by the same idea as in Lemma 4.5 to prove this (see Figure 10). Consider the “half”-tilted grim reaper cylinder

$$\mathcal{G}_+^{t, \alpha - \epsilon} := \{x \in \mathcal{G}^{0, \alpha - \epsilon} : \langle x, e_1 \rangle \geq 0\} + te_1, \quad (22)$$

where  $\epsilon > 0$  and  $t \in [0, \infty)$ . Take a sufficiently large  $t_0$  so that

$$\mathcal{G}_+^{t_0, \alpha - \epsilon} \cap \text{spt } M_\infty = \emptyset.$$

By Lemma 4.5 this is possible. Consider the set

$$\mathcal{A} = \{t \in [0, +\infty) : \mathcal{G}_+^{t, \alpha - \epsilon} \cap \text{spt } M_\infty = \emptyset\},$$

which is nonempty. We will show that  $\inf \mathcal{A} = 0$ . Indeed, otherwise,  $s_0 = \inf \mathcal{A} > 0$  satisfy one of the following conditions:

- a.  $\mathcal{G}_+^{s_0, \alpha - \epsilon}$  and  $\text{spt } M_\infty$  have a point of contact;
- b.  $\text{dist}(\mathcal{G}_+^{s_0, \alpha - \epsilon}, \text{spt } M_\infty) = 0$ .

According to Theorem 2.3 and Lemma 4.5, the 1st case is not possible. Regarding the 2nd case, by Lemma 4.5 there exists a sequence  $\{z_i = (z_i^1, \dots, z_i^{n+1})\}$  in  $\text{spt } M_\infty$  such that

- i.  $\lim_i \text{dist}(z_i, \mathcal{G}_+^{s_0, \alpha - \epsilon}) = 0$ ;
- ii.  $\{z_i^1\} \rightarrow z_\infty^1$  and  $a < z_i^1 - t < b$ , where  $0 < a < b < \frac{\pi}{2 \cos(\theta)}$  are constants;
- iii.  $\{\langle z_i, u_\theta \rangle\} \rightarrow z_\infty^{u_\theta}$ ;
- iv. the sequence  $\{(0, z_i^2, \dots, z_i^{n+1}) - \langle z_i, u_\theta \rangle u_\theta\}$  is unbounded;
- v.  $\Theta(\text{spt } M_\infty, z_i) \geq 1$ , where  $\Theta(\text{spt } M_\infty, \cdot)$  is the density of the Radon measure associated to  $M_\infty$ .

At this point, let us consider the sequence

$$\{\mathfrak{M}_i := M_\infty - (0, z_i^2, \dots, z_i^{n+1}) + \langle z_i, u_\theta \rangle u_\theta\}.$$

By Remark 3.1 we can suppose  $\mathfrak{M}_i \rightarrow \mathfrak{M}_\infty$ , where  $\mathfrak{M}_\infty$  is a connected stationary integral varifold. Following the same arguments that we used to show that  $p_\infty \in \text{spt } M_\infty$  in Lemma 3.2, we see that

$$z_\infty^1 \mathbf{e}_1 + z_\infty^{u_\theta} u_\theta \in \text{spt } \mathfrak{M}_\infty \cap \mathcal{G}_+^{s_0, \alpha - \epsilon}.$$

Moreover, note that item ii implies that  $z_\infty^1 \mathbf{e}_1 + z_\infty^{u_\theta} u_\theta$  is an interior point of  $\mathcal{G}_+^{s_0, \alpha - \epsilon}$ . Therefore, by Theorem 2.3 and Lemma 4.5 we obtain a contraction. Thus,  $\inf \mathcal{A} = 0$  and

$$\mathcal{G}_+^{0, \alpha - \epsilon} \cap \text{spt } M_\infty = \emptyset$$

because  $\epsilon > 0$  and  $\inf_{\text{spt } M_\infty} \langle x, u_\theta \rangle = \alpha$ . Similarly, we deduce that

$$\mathcal{G}_-^{0, \alpha - \epsilon} \cap \text{spt } M_\infty = \emptyset.$$

Hence,  $\mathcal{G}^{0, \alpha - \epsilon} \cap \text{spt } M_\infty = \emptyset$ .

Now taking  $\epsilon \rightarrow 0^+$  and using the fact that

$$\inf_{\text{spt } M_\infty} \langle x, u_\theta \rangle = \min_{\text{spt } M_\infty} \langle x, u_\theta \rangle = \alpha$$

we conclude that  $\text{spt } M_\infty$  touches the tilted grim reaper cylinder  $\mathcal{G}^{0, \alpha}$  at  $p_\infty^1 \mathbf{e}_1 + \alpha u_\theta$ . In particular, by Theorem 2.3 we conclude that  $M_\infty = \mathcal{G}^{0, \alpha}$ . This concludes the proof of our claim. Moreover, by Theorem 2.7 we have  $M_i \rightarrow M_\infty = \mathcal{G}^{0, \alpha}$ , with multiplicity 1.

Let us consider the sets

$$S_i := \{v \in \mathbb{R}^{n+1} : \langle v, E_n \rangle \leq -i\},$$

where  $i \in \mathbb{N}$ . Take  $\beta = \lim_i \inf_{S_i \cap M} \langle x, u_\theta \rangle$ . Let  $\{q_i = (q_i^1, \dots, q_i^{n+1})\}$  be a sequence in  $M$  such that

- i.  $q_i \in S_i \cap M$  and  $\langle q_i, u_\theta \rangle - \inf_{S_i \cap M} \langle x, u_\theta \rangle < \frac{1}{i}$ ;
- ii.  $\{q_i^1\} \rightarrow q_\infty^1$  and  $-\frac{\pi}{2 \cos(\theta)} < q_\infty^1 < \frac{\pi}{2 \cos(\theta)}$ ;
- iii.  $\{\langle q_i, u_\theta \rangle\} \rightarrow \beta$ .

Then, reasoning as before, we obtain

$$N_i := M - (0, q_i^2, \dots, q_i^{n+1}) + \langle q_i, u_\theta \rangle u_\theta \longrightarrow \mathcal{G}^{0, \beta},$$

with multiplicity 1.

By Lemma 4.4, we know that there exists a sufficiently large  $t_0$ , so that  $M^+(t_0)$  is a graph over an open set in the hyperplane  $[\mathbf{e}_{n+1}]^\perp$ . Moreover, we can choose at the same time a small enough  $\tau > 0$  so that

$$M_+ \left( \frac{\pi}{2 \cos(\theta)} - 2\tau \right) \cup M_+ \left( -\frac{\pi}{2 \cos(\theta)} + 2\tau \right) \subset M^+(t_0).$$

Hence, there is  $i_0 \in \mathbb{N}$  such that

- a. there exist strictly increasing sequences of positive numbers  $\{m_i^1\}, \{m_i^2\}, \{n_i^1\}$ , and  $\{n_i^2\}$  so that

$$m_i^1 < m_i^2 \text{ and } -n_i^1 < -n_i^2, \text{ for all } i > i_0;$$

- b. there exist smooth functions

$$\varphi_i : \left( -\frac{\pi}{2 \cos(\theta)} + \tau, \frac{\pi}{2 \cos(\theta)} - \tau \right) \times (m_i^1, m_i^2)^{n-1} \longrightarrow \mathbb{R} \quad (23)$$

and

$$\phi_i : \left( -\frac{\pi}{2 \cos(\theta)} + \tau, \frac{\pi}{2 \cos(\theta)} - \tau \right) \times (-n_i^1, -n_i^2)^{n-1} \longrightarrow \mathbb{R} \quad (24)$$

satisfying

$$|\varphi_i| < \frac{1}{i}, |D\varphi_i| < \frac{1}{i}, |\phi_i| < \frac{1}{i} \text{ and } |D\phi_i| < \frac{1}{i}, \text{ for all } i > i_0 \quad (25)$$

and such that the hypersurfaces

$$R_i := \left\{ (x_1, \dots, x_{n+1}) \in M : -\frac{\pi}{2 \cos(\theta)} + \tau < x_1 < \frac{\pi}{2 \cos(\theta)} - \tau \right. \\ \left. (x_2, \dots, x_{n-1}) \in (m_i^1, m_i^2)^{n-2}, \langle x, E_n \rangle \in (m_i^1, m_i^2) \right\}$$

and

$$L_i := \left\{ (x_1, \dots, x_{n+1}) \in M : -\frac{\pi}{2\cos(\theta)} + \tau < x_1 < \frac{\pi}{2\cos(\theta)} - \tau \right. \\ \left. (x_2, \dots, x_{n-1}) \in (-n_i^1, -n_i^2)^{n-2}, \langle x, E_n \rangle \in (-n_i^1, -n_i^2) \right\}$$

can be written as graphs of functions  $\varphi_i$  and  $\phi_i$ , respectively, over the corresponding pieces of the tilted grim reaper cylinder as in the proof of Lemma 4.4, where  $E_n := \cos(\theta)\mathbf{e}_n + \sin(\theta)\mathbf{e}_{n+1}$ .

Now following the same idea as in Lemma 4.4, we see that  $R_i$  and  $L_i$  are graphs over domains in the hyperplane  $[\mathbf{e}_{n+1}]^\perp$  (for  $i_0$  large enough). Note that  $R_i$  and  $L_i$  are connected because they are graphs over the connected sets and the convergence has multiplicity 1. Finally, let us consider the exhaustion  $\{\Lambda_i\}$  of  $M$  by compacts sets given by

$$\Lambda_i := \left\{ x = (x_1, \dots, x_{n+1}) \in M : (x_2, \dots, x_{n-1}) \in [-a_i, b_i]^{n-2} \right. \\ \left. \langle x, E_n \rangle \in [-a_i, b_i] : \langle x, u_\theta \rangle \leq i \right\}, \quad (26)$$

where  $a_i = \frac{n_i^1 + n_i^2}{2}$  and  $b_i = \frac{m_i^1 + m_i^2}{2}$ .

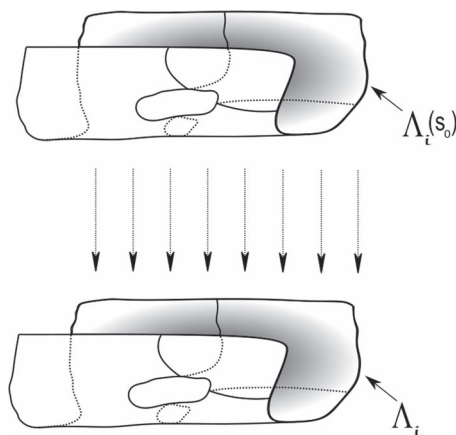
Since  $M^+(t_0)$ ,  $R_i$ , and  $L_i$  are vertical graphs, then a small strip  $B_i$  around the boundary of  $\Lambda_i$  is a graph over the hyperplane  $[\mathbf{e}_{n+1}]^\perp$ . Using the Rado's classical argument, the former fact implies that  $\Lambda_i$  is a graph over the hyperplane  $[\mathbf{e}_{n+1}]^\perp$  if  $i > i_0$ . Indeed, assume to the contrary that this is not true. Consider the family

$$\{\Lambda_i(s) := \Lambda_i + s\mathbf{e}_{n+1}\}_{s \in \mathbb{R}}$$

of translations of  $\Lambda_i$  into the direction of  $\mathbf{e}_{n+1}$ . Since  $\Lambda_i$  is compact there exists a sufficiently large  $s_0$  so that

$$\Lambda_i(s_0) \cap \Lambda_i = \emptyset.$$

Now move  $\Lambda_i(s_0)$  into direction of  $-\mathbf{e}_{n+1}$  (see Figure 11). Since  $\Lambda_i$  is not a graph and  $B_i \cap \{B_i + s\mathbf{e}_{n+1}\} = \emptyset$ , because  $B_i$  is a graph over a subset of  $[\mathbf{e}_{n+1}]^\perp$ . Then there exists an  $s_1 \in (0, s_0)$  such that  $\Lambda_i(s_1)$  has a point of contact at interior with  $\Lambda_i$ . Therefore,  $\Lambda_i(s_1) = \Lambda_i$ , but this gives us a contraction. Hence, each  $\Lambda_i$  must be a graph. Since  $\bigcup_i \Lambda_i = M$ , then  $M$  is also a vertical graph. ■



**Fig. 11.** Picture of  $\Lambda_i$  and  $\Lambda_i(s_0)$ .

Observe that the previous lemma implies that  $H > 0$ . Hence, given any  $v \in \mathbb{R}^{n+1}$  if  $\xi_v = \langle \xi, v \rangle$ , then  $h_v = \frac{\xi_v}{H}$  are well defined on  $M$ , where  $\xi$  stands for the Gauss map of  $M$ , and  $H$  is the mean curvature of  $M$ .

**Lemma 4.7.** The functions  $\xi_v$ ,  $H$ , and  $h_v = \frac{\xi_v}{H}$  satisfy the following equalities:

$$\Delta \xi_v + \langle \nabla \xi_v, \nabla x_{n+1} \rangle + |A|^2 \xi_v = 0, \quad (27)$$

$$\Delta H + \langle \nabla H, \nabla x_{n+1} \rangle + |A|^2 H = 0, \quad (28)$$

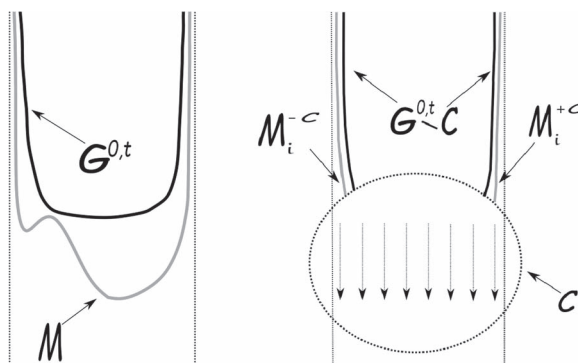
$$\Delta h_v + \langle \nabla h_v, \nabla (x_{n+1} + 2 \log H) \rangle = 0. \quad (29)$$

**Proof.** Equation (28) was proved in [11, Lemma 2.1]. Similarly, the proof of (27) can be found in [16, Equation (3.4.10)]. Equation (29) follows from the combination of (27) and (28). ■

Now we will show that our hypersurface is  $C^2$ -asymptotic to two half-hyperplanes with respect to the Euclidean metric.

**Lemma 4.8.** The hypersurface  $M$  is  $C^2$ -asymptotic outside the cylinder to two half-hyperplanes with respect to the Euclidean metric.

**Proof.** To prove this lemma, we will need the following claim.



**Fig. 12.** Transversal section of the behavior of  $M$  and  $\mathcal{G}^{0,t}$  in the left side and transversal section of the behavior of  $M \setminus C$  and  $\mathcal{G}^{0,t} \setminus C$  in the right side.

**Claim 4.8.1.** There exists a tilted grim reaper cylinder inside the region that lies above  $M$ .

*Proof of claim:* Note that if  $t$  is sufficiently large then  $M^+(t)$  is a graph over the hyperplane  $\Pi(0)$ . Now take the tilted grim reaper  $\mathcal{G}^{0,t}$ . We will show that it lies in the region above  $M$ . As in Lemma 4.5, let us consider the family

$$\{M_*(s) := M_+(0) + s\mathbf{e}_1\}_{s \in [0, +\infty)}.$$

Taking into account the asymptotic behavior of  $M$ , there exists a sufficiently large  $s_0 > 0$  so that  $M_*(s_0) \cap \mathcal{G}^{0,t} = \emptyset$ . Applying the same argument as in Lemma 4.5 and the fact that  $M^+(t)$  is a graph over  $\Pi(0)$ , we conclude that  $\inf \mathcal{A} = 0$ , where

$$\mathcal{A} := \{s \in [0, +\infty) : M_*(s) \cap \mathcal{G}^{0,t} = \emptyset\}.$$

In particular, it holds that  $M_+(0) \cap \mathcal{G}^{0,t} = \emptyset$ . Applying the same argument to the family

$$\{M^*(s) := M_-(0) - s\mathbf{e}_1\}_{s \in [0, +\infty)},$$

we obtain  $M_-(0) \cap \mathcal{G}^{0,t} = \emptyset$ . Hence,  $M \cap \mathcal{G}^{0,t} = \emptyset$  and this proves the claim (see Figure 12).

Now take any cylinder  $C$  so that  $M \setminus C$  and  $\mathcal{G}^{0,t} \setminus C$  have two connected components, which are  $C^1$ -asymptotic to two half-hyperplanes. Clearly the family

$$\{\mathcal{G}_C^{0,t} := \mathcal{G}^{0,t} \setminus C - i \cdot u_\theta\}_{i \in \mathbb{N}}$$

converges with respect to the  $C^\infty$ -topology to two hyperplanes  $\Pi(-\frac{\pi}{2\cos(\theta)})$  and  $\Pi(\frac{\pi}{2\cos(\theta)})$ . This forces the family  $\{M_i^C := M \setminus \mathcal{C} - i \cdot u_\theta\}_{i \in \mathbb{N}}$  to converge as sets to these hyperplanes (see Figure 12). By Theorem 2.6 each component of  $M_i^C$  is stable. Let  $M_i^{-C}$  and  $M_i^{+C}$  be the two connected components of  $M_i^C$  that lie on the left side of the hyperplane  $\Pi(0)$  and on the right side of the hyperplane  $\Pi(0)$ , respectively. It is important to observe that

$$\text{Area}(M_i \cap K) \leq C(K)$$

for every compact set  $K \subset \mathbb{R}^{n+1}$ , where  $C(K)$  is a constant only depending on  $K$  and the dimension  $n$ , and the area is to take with respect Ilmanen's metric. Therefore, by Theorem 2.5, the sequences  $\{M_i^{-C}\}$  and  $\{M_i^{+C}\}$  converge in the  $C^\infty$ -topology (with multiplicity 1) to the hyperplanes  $\Pi(-\frac{\pi}{2\cos(\theta)})$  and  $\Pi(\frac{\pi}{2\cos(\theta)})$ , respectively, away from the singular set, with respect to the Ilmanen's metric. Now by Theorem 2.7, we obtain that the convergence is  $C^\infty$  with multiplicity 1 everywhere. This proves that  $M$  is  $C^\infty$ -asymptotic to the half-hyperplanes of  $\Pi(-\frac{\pi}{2\cos(\theta)})$  and  $\Pi(\frac{\pi}{2\cos(\theta)})$ , with respect to the Ilmanen's metric.

Finally, we claim that  $M$  is  $C^2$ -asymptotic to the half-hyperplanes  $\Pi(-\frac{\pi}{2\cos(\theta)})$  and  $\Pi(\frac{\pi}{2\cos(\theta)})$ . Let us work with the wing of  $M$  that is  $C^1$ -close to the half-hyperplane  $\mathcal{H}_1$  of  $\Pi(\frac{\pi}{2\cos(\theta)})$ . As we know, given  $\epsilon > 0$ , there exists  $\delta > 0$  so that  $M$  can represent a graph of  $\varphi$  defined over  $\mathcal{H}_1$ , with  $\sup_{\mathcal{H}_1(\delta)} |\varphi| < \epsilon$  and  $\sup_{\mathcal{H}_1(\delta)} |D\varphi| < \epsilon$ . Arguing by contraction, from the definition of  $C^2$ -asymptotic implies there exist  $\epsilon > 0$  and a sequence  $\{p_i\}$  in  $M$  such that

$$|D^2\varphi(p_i)| \geq \epsilon \text{ and } \langle p_i, u_\theta \rangle \rightarrow \infty. \quad (30)$$

Consider the sequence  $\{M_i := M - p_i\}$ . The argument above shows that the wings of  $M_i$  that are asymptotic to  $\mathcal{H}_1 - p_i$  converge to  $\Pi(0)$  in the  $C^\infty$ -topology, with respect to Ilmanen's metric. Notice that this wing is a graph of a function  $\varphi_i(\cdot) = \varphi(\cdot + p_i) - \varphi(p_i)$  that is defined in a half-hyperplane of  $\Pi(0)$  that contains the origin.

Consider a small geodesic cylinder  $W_{r,\epsilon}$  around  $0 \in \mathbb{R}^{n+1}$ , with respect to Ilmanen's metric. By definition of convergence in the  $C^\infty$ -topology, there exists sufficiently large  $i_0 \in \mathbb{N}$  so that for all  $i > i_0$  the set  $W_{r,\epsilon} \cap M_i$  is a graph of a function  $\eta_i$  defined over  $B_r(p) \subset \Pi(0)$  such that  $\sup_{B_r \cap \Pi(0)} |D^l \eta_i| < \epsilon/8$ , for all  $l \in \mathbb{N}$ . Notice that the hyperplanes parallel to  $\mathbf{e}_{n+1}$  are totally geodesic, and  $\mathbf{e}_1$  is normal vector to  $T_0 \Pi(0)$ , and we have the following relation between  $\varphi_i$  and  $\eta_i$ :

$$\varphi_i(\exp_0(q + \eta_i(q)\mathbf{e}_1)) - \langle \exp_0(q + \eta_i(q)\mathbf{e}_1), \mathbf{e}_1 \rangle \mathbf{e}_1) = \langle \exp_0(q + \eta_i(q)\mathbf{e}_1), \mathbf{e}_1 \rangle.$$

Differentiating twice and evaluating at  $q = 0$ , we deduce that

$$\begin{aligned} \langle D^2 \exp_0(\bar{u}_i, \bar{w}_i), \mathbf{e}_1 \rangle + D^2 \eta_i(u, w) &= D^2 \varphi_i(u, w) \\ &+ d\varphi_i[D^2 \exp_0(\bar{u}_i, \bar{w}_i) - \langle D^2 \exp_0(\bar{u}_i, \bar{w}_i), \mathbf{e}_1 \rangle \mathbf{e}_1], \end{aligned}$$

where  $\bar{u}_i := u + d\eta_i(u)\mathbf{e}_1$ . From this expression, the control on the  $C^\infty$  norm of  $\eta_i$ , and on the  $C^1$  norm of  $\varphi$  we get a contraction with  $|D^2 \varphi_i(0)| = |D^2 \varphi(p_i)| \geq \epsilon$ , if  $i$  is sufficiently large. This proves the lemma.  $\blacksquare$

For the sake of simplicity, let us set  $h_j := \frac{\langle \xi, \mathbf{e}_j \rangle}{H}$ , where  $j \in \{2, \dots, n-1\}$  and  $h_n = \frac{\langle \xi, E_n \rangle}{H}$  (recall that  $E_n = \cos(\theta)\mathbf{e}_n + \sin(\theta)\mathbf{e}_{n+1}$ ). Using the previous lemma we can obtain some information about the behavior of the functions  $h_j$  at the ends of  $M$ .

**Lemma 4.9.** The functions  $h_j, j \in \{2, \dots, n\}$  tend to zero as we approach the end of  $M$ .

**Proof.** Consider the exhaustion  $\{\Lambda_i\}$  given by (26). Notice that the boundary of each  $\Lambda_i$  consists of the following  $2n-1$  regions:

$$\begin{aligned} \Lambda_i^1 &:= \left\{ x = (x_1, \dots, x_{n+1}) \in M : (x_2, \dots, x_{n-1}) \in [-a_i, b_i]^{n-2} \right. \\ &\quad \left. \langle x, E_n \rangle \in [-a_i, b_i], \langle x, u_\theta \rangle = i \right\} \\ \Lambda_i^{-2} &:= \{x = (x_1, \dots, x_{n+1}) \in M : (x_3, \dots, x_{n-1}) \in [-a_i, b_i]^{n-3}, \langle x, u_\theta \rangle \leq i \\ &\quad x_2 = -a_i, \langle x, E_n \rangle \in [-a_i, b_i]\} \\ \Lambda_i^{+2} &:= \{x = (x_1, \dots, x_{n+1}) \in M : (x_3, \dots, x_{n-1}) \in [-a_i, b_i]^{n-3}, \langle x, u_\theta \rangle \leq i \\ &\quad x_2 = b_i, \langle x, E_n \rangle \in [-a_i, b_i]\} \\ &\vdots \\ \Lambda_i^{-n} &:= \{x = (x_1, \dots, x_{n+1}) \in M : (x_2, \dots, x_{n-1}) \in [-a_i, b_i]^{n-2} \\ &\quad \langle x, u_\theta \rangle \leq i, \langle x, E_n \rangle = -a_i\} \end{aligned}$$

and

$$\begin{aligned} \Lambda_i^{+n} &:= \{x = (x_1, \dots, x_{n+1}) \in M : (x_2, \dots, x_{n-1}) \in [-a_i, b_i]^{n-2} \\ &\quad \langle x, u_\theta \rangle \leq i, \langle x, E_n \rangle = b_i\}. \end{aligned}$$

Let us study the behavior of  $h_j$  in a small strip around the boundary of  $\Lambda_i$ . We begin our analysis in the connected component  $\Lambda_i^1$ . Fix any sufficiently small  $\epsilon > 0$ . Taking into account Lemma 4.8 and the definition of  $M^+(t)$ , we can use a similar argument as in the proof of Lemma 4.4 to guarantee the existence of a sufficiently large  $i_1 (> i_0)$ , a sufficiently small  $\tau > 0$ , and a smooth function  $\varphi$  defined on the strip

$$\mathcal{S}_\tau := \left[ \left( -\frac{\pi}{2\cos(\theta)}, -\frac{\pi}{2\cos(\theta)} + \tau \right) \cup \left( \frac{\pi}{2\cos(\theta)} - \tau, \frac{\pi}{2\cos(\theta)} \right) \right] \times \mathbb{R}^{n-1}$$

satisfying

$$\sup_{\mathcal{S}_\tau} |\varphi| < \epsilon, \quad \sup_{\mathcal{S}_\tau} |D\varphi| < \epsilon, \quad \text{and} \quad \sup_{\mathcal{S}_\tau} |D^2\varphi| < \epsilon \quad (31)$$

and such that  $M^+(i_1)$  is a graph of this function over the corresponding strip in the tilted grim reaper cylinder. From (14) and (19) we obtain

$$h_j = \frac{\alpha(j)}{\cos(\theta)} \frac{\partial_{x_j} \varphi}{\cos(x_1 \cos(\theta))} \frac{1 + \varphi \cos(\theta) \cos(x_1 \cos(\theta))}{1 + \sigma(\varphi, D\varphi)}, \quad (32)$$

where  $\alpha(j) = (-1)^j \left\lceil n - \frac{j+1}{2} \right\rceil$ , if  $j \in \{2, \dots, n-1\}$  and  $\alpha(n) = (-1)^{n-1} \cos(\theta)$ . Here

$$\begin{aligned} \sigma(\varphi, D\varphi) &:= \varphi \cos(\theta) \cos(x_1 \cos(\theta)) \\ &+ (-1)^n \{ \sin(\theta) (1 + \varphi \cos(\theta)) \partial_{x_n} \varphi + \sin(x_1 \cos(\theta)) \partial_{x_1} \varphi \}. \end{aligned}$$

Using the fact that  $M^+(i_1)$  is a graph over the tilted grim reaper cylinder and it is  $C^2$ -asymptotic to the half-hyperplane, we conclude that for all fixed  $(x_2, \dots, x_n)$  we have

$$\lim_{x_1 \rightarrow \frac{\pi}{2\cos(\theta)}^-} |\varphi| = \lim_{x_1 \rightarrow \frac{\pi}{2\cos(\theta)}^-} |D\varphi| = 0.$$

Therefore,

$$|\partial_{x_j} \varphi(x_1, x_2, \dots, x_n)| = \left| - \int_{x_1}^{\frac{\pi}{2\cos(\theta)}} \partial_{x_j x_1} \varphi(x, x_2, \dots, x_n) dx \right| \leq \left( \frac{\pi}{2\cos(\theta)} - x_1 \right) \epsilon. \quad (33)$$

So, by (31)–(33) we obtain that  $|h_j(x)| < o(\epsilon)$ , for all  $x$  near  $\Lambda_i^1$ . Thus,

$$\sup_{N(\Lambda_i^1)} |h_j| < o(\epsilon), \quad (34)$$

where  $N(\Lambda_i^1)$  is a small neighborhood of the  $\Lambda_i^1$  in  $\Lambda_i$ , if  $i > i_1$ .

Now we are going to work with the components of  $\partial\Lambda_i$  that intersect  $M^-(i_1)$ . Since  $R_i$  and  $L_i$  are  $C^1$ -close to a strip in the tilted grim reaper cylinder, there is sufficiently large  $i_2$  such that  $R_i \cap \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1}; \langle x, u_\theta \rangle \leq i_1\}$  is a graph over the strip in the tilted grim reaper cylinder of a function  $\varphi_i$  defined in the strip

$$G_\tau := \left( -\frac{\pi}{\cos(\theta)2} + \frac{\tau}{2}, \frac{\pi}{2\cos(\theta)} - \frac{\tau}{2} \right) \times (m_i^1, m_i^2)^{n-1}$$

satisfying the following properties:

$$\sup_{G_\tau} |\varphi_i| < \epsilon \text{ and } \sup_{G_\tau} |D\varphi_i| < \epsilon. \quad (35)$$

The same estimate is true for  $L_i$ . Furthermore, since  $\cos(x_1 \cos(\theta)) > \kappa > 0$  in  $G_\tau$ , for a suitable constant  $\kappa$ , then (35) and (32) give us that  $\sup_{G_\tau} |h_j| < o(\epsilon)$ . Hence,

$$\sup_{N(\Lambda_i^{\pm k})} |h_j| < o(\epsilon), \quad (36)$$

where  $k \in \{2, \dots, n\}$  and  $N(\Lambda_i^{\pm k})$  is a small neighborhood of the  $\Lambda_i^{\pm k}$  in  $\Lambda_i$ . Hence, for (34) and (36) we have  $\sup_{N(\partial\Lambda_i)} |h_j| < o(\epsilon)$ , for any  $i \in \mathbb{N}$ ,  $i > \max\{i_1, i_2\}$ . ■

Now we are ready to prove the main theorem of this paper.

**Proof of Theorem 4.1.** Recall that we are assuming that  $M$  is asymptotic to half-hyperplanes that are contained in different hyperplanes and that  $\inf_M \langle x, u_\theta \rangle = 0$ . According to Lemma 4.9 there is an interior point where  $h_j$  has an extremum. Then because  $h_j$  is a solution of (29), we can apply Hopf's maximum principle to conclude that  $h_j = 0$ , that is,  $\xi_j = 0$  on  $M$  for all  $j \in \{2, \dots, n\}$ . In particular, each  $e_j$  and  $E_n$  are tangent vectors of  $M$  for  $j \in \{2, \dots, n-1\}$  at all point of  $M$ . Thus, we can consider a global orthonormal frame in  $M$ ,  $\{E_1, E_j = e_j; j \in \{2, \dots, n-1\}; E_n\}$  (see (13)), where  $E_1 = E_2 \wedge \dots \wedge E_n \wedge \xi$ . Differentiating each  $\xi_j, j \in \{2, \dots, n\}$ , with respect to  $E_k, k \in \{1, \dots, n\}$ , we deduce

$$0 = E_k(\xi_j) = E_k \langle \xi, E_j \rangle = \langle D_{E_k} \xi, E_j \rangle = A(E_k, E_j).$$

Thus,

$$|A|^2 = \sum_{i,j} A(E_j, E_k)^2 = A(E_1, E_1)^2 = H^2.$$

Therefore, by [11, Theorem B] we conclude that  $M = \mathcal{G}^{0,0}$ , because  $0 = \inf_M \langle x, u_\theta \rangle$ . ■

#### 4.2 The case $\theta = \pi/2$

Now we are going to work in the case when the cylinder is vertical, that is, the axis of the cylinder is parallel to the translating velocity. The philosophy now is to compare  $M$  with a suitable translation of itself to arrive at a contraction. Then we need that the limit of the translated hypersurfaces is also smooth, but this holds true if  $n < 7$ . However, we conjecture that the result is true for any dimension. First of all, we would like to point out the following.

**Remark 4.1.** The Dynamics lemma (Lemma 3.1) is still true in this situation, for every  $n \geq 2$ . The proof works exactly as in the case  $\theta < \pi/2$ , except the proof that the sequence  $\{M_t^-(\delta)\}$  has locally bounded area. In this case, in order to prove that the area blow-up set is empty we use as barriers the family  $P_\lambda = W_\lambda^2 \times \mathbb{R}^{n-2}$  (cylinders over the translating catenoid of dimension 2), for a sufficiently large  $\lambda > 0$  so that the cylinder lies inside the neck of  $P_\lambda = W_\lambda^2 \times \mathbb{R}^{n-2}$ . Hence, if the set of area blow-up is not empty, then we could move  $P_\lambda = W_\lambda^2 \times \mathbb{R}^{n-2}$  until we get a 1st contact point with the area blow-up set, which is impossible.

**Theorem 4.2** (Vertical Cylinders). Let  $M^n \subset \mathbb{R}^{n+1}$  be a complete, connected, properly embedded translating soliton, and  $\mathcal{C} := \{x \in \mathbb{R}^{n+1} : \langle x, \mathbf{e}_1 \rangle^2 + \langle x, \mathbf{e}_n \rangle^2 \leq r^2\}$ , for  $r > 0$ . Assume that  $M$  is  $C^1$ -asymptotic to two half-hyperplanes outside  $\mathcal{C}$  and  $n < 7$ . Then  $M$  must coincide with a hyperplane parallel to  $\mathbf{e}_{n+1}$ .

**Proof.** We claim that  $\mathcal{H}_1$  and  $\mathcal{H}_2$  are parallel. Assume to the contrary that is true. Then we could take a hyperplane parallel to  $\mathbf{e}_{n+1}$ ,  $\Gamma$ , such that it does not intersect  $M$  and such that the normal vector  $v$  to  $\Gamma$  is not perpendicular to  $w_1$  and  $w_2$ . Translate  $\Gamma$  by  $t_0 \in \mathbb{R}$  in the direction of  $v$  until we get a hyperplane  $\Gamma_{t_0} := \Gamma + t_0 v$  such that either  $\Gamma_{t_0}$  and  $M$  have a 1st point of contact or  $\text{dist}(\Gamma_{t_0}, M) = 0$  and  $\Gamma_{t_0} \cap M = \emptyset$ . The 1st case is not possible by Theorem 2.3. Regarding the 2nd case, we can reason as in Lemma 3.2 to see that this case is also impossible.

Notice that we cannot have either  $\mathcal{H}_1 \subset \mathcal{H}_2$  or  $\mathcal{H}_2 \subset \mathcal{H}_1$  because in these cases we could take a hyperplane parallel to  $\mathbf{e}_{n+1}$ ,  $\Upsilon$ , whose normal is exactly  $w_1$  and do not intersect  $M$ . Now we could move  $\Upsilon$  into direction of  $w_1$  until there exists  $t_0 > 0$  such that either  $\Upsilon + t_0 w_1$  and  $M$  have a 1st point of contact or  $\{\Upsilon + t_0 w_1\} \cap M = \emptyset$  and  $\text{dist}(\Upsilon + t_0 w_1, M) = 0$ . Reasoning as in the above paragraph, we can conclude that both situations are impossible.

**Remark 4.2.** We would like to point out that until this point the argument is valid for any dimension. In the remaining part of the proof we shall need that  $n < 7$ .

Denote by  $\Pi_1$  and  $\Pi_2$  the hyperplanes that contain the half-hyperplane  $\mathcal{H}_1$  and  $\mathcal{H}_2$ , respectively. By the previous claims,  $\Pi_1$  and  $\Pi_2$  are parallel. We claim that  $\Pi_1$  and  $\Pi_2$  coincide. Notice that this proves our theorem because if we reason as at the end of Lemma 3.2 we can deduce that  $M$  coincides  $\Pi_1 (= \Pi_2)$ .

Suppose to the contrary that  $\Pi_1 \neq \Pi_2$ . Let  $\nu$  be the normal vector to  $\Pi_1$ , and take  $s_0$  sufficiently large so that  $M + s_0\nu$  does not intersect the slab limited by  $\Pi_1$  and  $\Pi_2$ . Now consider a sufficiently large  $t_0 > 0$  so that  $(M \cap \mathcal{C}) + s_0\nu + t_0w_1$  lies in  $\mathcal{Z}_{1,\delta}^+$  (see Lemma 3.1) and notice that the wing of  $M$  that corresponds to  $\mathcal{H}_1(\delta)$  is a graph over this set. Define the set

$$\mathcal{A} := \{s \in [0, \infty); \{M + s\nu + t_0w_1\} \cap M = \emptyset\}.$$

Let  $s_1 := \inf \mathcal{A} > 0$ . We have two possibilities: either  $s_1 \notin \mathcal{A}$  or  $s_1 \in \mathcal{A}$ . The 1st case implies that  $M + s_1\nu + t_0w_1$  and  $M$  have a point of contact, which is impossible by Theorem 2.3. In the 2nd case, we must have

$$\text{dist}(M + s_1\nu + t_0w_1, M) = 0$$

and  $\{M + s_1\nu + t_0w_1\} \cap M = \emptyset$ . This means that there exist sequences  $\{p_i\}$  in  $M \cap \mathcal{Z}_{1,\delta}^+$  and  $\{q_i\}$  in  $\{M + s_1\nu + t_0w_1\} \cap \mathcal{Z}_{1,\delta}^+$  such that  $\text{dist}\{p_i, q_i\} = 0$ , note that we can suppose  $\{\langle q_i, \mathbf{e}_1 \rangle\}, \{\langle p_i, \mathbf{e}_1 \rangle\} \rightarrow a$  and  $\{\langle q_i, \mathbf{e}_n \rangle\}, \{\langle p_i, \mathbf{e}_n \rangle\} \rightarrow b$ . Consider the sequences

$$\{M_i := M - (0, p_2, \dots, p_{n-1}, 0, p_{n+1})\}$$

and

$$\{\widehat{M}_i := M + s_1\nu + t_0w_1 - (0, q_2, \dots, q_{n-1}, 0, q_{n+1})\}.$$

By Remark 4.1, up to a subsequence,  $M_i \rightarrow M_\infty$  and  $\widehat{M}_i \rightarrow \widehat{M}_\infty$ , where  $M_\infty$  and  $\widehat{M}_\infty$  are connected stationary integral varifold and  $(a, 0, \dots, 0, b, 0) \in \text{spt} M_\infty \cap \text{spt} \widehat{M}_\infty$ . Furthermore, by Theorem 2.5 and our hypothesis, the point  $(a, 0, \dots, 0, b, 0)$  lies in the region, where  $M_\infty$  is smooth. Hence, Theorem 2.3 implies that  $M_\infty = \widehat{M}_\infty$ , which is impossible. Therefore,  $\Pi_1 = \Pi_2 = M$ . ■

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